The Efficacy of Virtual Laboratories as Undergraduate Educational Resources

Patrick Coleman – November 2021

This work was submitted as a component of the Bachelor of Advanced Studies (Honours) course for the School of Life and Environmental Sciences at the University of Sydney.

Declaration of originality

All content within this thesis is entirely the product of the candidate's own research for the University of Sydney and no part of this work has been submitted for a degree or qualification to another university of institution.

Signed: Patrick

Acknowledgements

Special thanks go to Dr Matthew Clemson and Prof Gareth Denyer as co-supervisors overseeing this project. Additional thanks to Dr Tara Christie, Dr Giselle Yeo, Dr Markus Hofer and Dr Lifeng Peng for the inclusion of the virtual laboratories in each of their courses.

Abstract

In 2020, most undergraduate courses quickly moved to an online delivery due to social distancing requirements imposed by governments in response to COVID-19. This presented challenges for educators to deliver content online, with practical components being particularly effected. Many previously face-to-face laboratories were replaced by recorded videos of laboratory protocols that were not considered to be engaging by students, and the change was detrimental to their educational experience. To address this problem, virtual laboratories were identified as a potential solution, allowing students to conduct the laboratory protocol in an online environment. Many available online virtual laboratory tools allow users to follow the steps of a protocol but are not sufficiently complex to allow them to deviate from it, and to experience the consequences of deviations from the defined experimental steps. Additionally, virtual laboratories available to course coordinators are not course-specific, and do not allow their students to obtain results pertaining to their unit of study and experimental context.

To address this, an interactive, data-generating virtual laboratory was developed to provide an active learning environment within which students could conduct their own experiments. Online virtual laboratories were created for three separate undergraduate biochemistry courses, two at the University of Sydney and one at the University of Wellington. These laboratories gave students the necessary tools to complete their experiments in a virtual environment and to have the capability to faithfully reproduce the experimental data that would result from their actions and decisions within the program. This would allow students to experiment in the program and troubleshoot the results they obtained, engaging in educational cycles of predication, action, and feedback.

The success of the program was measured through feedback from both students and educators after their interactions with the virtual laboratory in the context of their courses. The feedback collected indicated that both groups endorsed the further use of virtual laboratories as educational tools in undergraduate coursework.

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1 Introduction

1.1 The importance of practical laboratory components in higher education.

Practical laboratory work is a crucial component of any undergraduate biochemical course, providing students with hands-on experience involving the techniques most commonly used in modern research. This allows students to complete their qualification with the technical knowledge and physical skills required for future laboratory work. In addition to this physical component, practical coursework also aims to deepen the students' theoretical scientific understanding of the processes involved, and contextualise this knowledge by linking to the practical[2]. These goals have been previously summarised as the following can be summarised into three key outcomes for students: (1) understanding scientific concepts, (2) interest and motivation, and (3) scientific practical skills and problem-solving abilities[3] .

These key principles will be taken into consideration when assessing the success of a given laboratory throughout this study.

1.2 The rise of online learning

In recent years there has been a significant rise in the implementation of purely online laboratory experiences for students, and the growth of the number of distance education students is one such influencing factor.

Online learning and distance education in universities has been rising slowly but consistently worldwide in recent years. In the US, enrolments in distance education have been rising steadily since 2001, even though total admissions have fluctuated[4], and at the beginning of 2021 there were over 8 million students enrolled exclusively in distance education[5]. Similar trends are observed in Australia, with enrolments in external undergraduate university courses doubling over the last two decades[6]. This

growth can be attributed to the advantages that distance education holds over face-to-face teaching, with the main benefits consisting primarily of the ability to study away from the locations of the university, the ability to study outside of designed course hours as required by the student's schedule, and the lower costs associated with distance educations degrees costs[7] [3].

The recent COVID-19 pandemic also caused a large exacerbation of this trend, with public health recommendations forcing many universities to implement strict social distancing guidelines at the beginning of 2020[8]. This led to a vast majority of undergraduate coursework being quickly transitioned to an online environment[10, 11], and universities forced to abruptly transition previously face-to-face courses into a format that was suitable to be delivered online.

1.3 The emergency online replacement for practical labs

This change was relatively simple for theoretical components of course material, where lectures could be recorded and accessed online by students, with minimal downsides. The challenge, however, came with the implementation of practical coursework online, where course providers had to design an adequate online substitute, while still meeting the previously detailed outcomes of a practical course as best as possible.

At the University of Sydney, the most common emergency teaching response to COVID-19 was to create video recordings of the laboratory procedures to be viewed during synchronous online classes with students in small groups. This consisted of pre-recorded demonstrations of the laboratory protocol as carried out by teaching staff, with pre-generated data assigned to students for data analysis, post-laboratory discussions and laboratory reports to be completed. While this teaching method broadly contained the same content as the face-to-face laboratory, it fostered a passive learning environment rather than an active one, which is not as effective a teaching method when considering student learning[12]. A student from an affected one of these classes described the experience as follows "It was supplemented by online

videos and learning which covered the practical content. But I missed the general sense of being in the lab and doing the tasks in person."[13]

While the video recordings and sample data sets were sufficient for providing the raw learning material to students, clearly the content lacked the ability to engage students in the same way that a physical laboratory experience did. The most effective approach would be to foster an active learning environment for students to engage with, as they had previously in face-to-face settings.

An active learning environment, or "Interactive Engagement Methods" have been defined as "those designed at least in part to promote conceptual understanding through interactive engagement of students in heads-on (always) and hands-on (usually) activities which yield immediate feedback through discussion with peers and/or instructors"[14]. A virtual laboratory has the capacity to provide an effective active learning environment for students, permitting engagement in cycles of prediction, action, feedback and troubleshooting in a manner that mimics the learning experience in a physical laboratory.

1.4 The current state of virtual laboratories

If an active learning environment is to be created for students that works effectively in an online learning environment, they must be able to have meaningful interactions with the online laboratory and receive corresponding feedback from these engagements. One such technology that has the potential to allow for these complex interactions is that of the virtual laboratory, a tool which seeks to replicate certain aspects of the physical laboratory experience in a virtual setting. Before the COVID-19 pandemic, virtual laboratories were mostly used as a training tool, allowing students to gain some experience before transitioning to a real lab. Since the pandemic, their widespread use has grown dramatically, and in many cases they are now used as a complete replacement for their face-to-face counterparts[15].

Virtual laboratories have previously been demonstrated to have the capacity to teach students theoretical course material just as effectively as their physical counterparts[16, 17], and additionally possess some benefits over them. In some face-to-face laboratories, constraints such as time and financial limitations can mean that students

are only able to perform parts of a full experimental procedure. For example, in some practical classes, students may have to work in groups, or only interact with a small part of the entire experimental protocol. This leads to individual students only having a limited influence on the outcome of the protocol, where they are not forced to test their 'conceptual understanding through interactive engagement' as previously described. These types of step-by-step laboratory classes are colloquially known as 'Cookbook laboratories'[18], and are recognised to be inferior methods of teaching concepts and practical skills than more investigative approaches[19]. In an online environment many of these constraints are non-existent, and students are able to conduct their own experiments by their own design if the virtual environment is sufficiently complex. Additional potential benefits of virtual laboratories include circumventing many safety concerns that may limit the scope of the student experience, such as protocols that involve hazardous materials or infectious or pathogenic organisms that require containment procedures. Ethical constraints are similarly reduced, since no living tissue is required in a virtual setting.

Conversely, there are also certain recognised limitations to virtual laboratories, where the full experience of a physical laboratory cannot be emulated. Regarding student experience, students do not rate their enjoyment of virtual laboratories as highly as physical laboratories, and find them less interesting overall[20]. It has also been found that students display minimal voluntary participation in virtual laboratories[21]. As student enjoyment is a large influencing factor on achieving learning outcomes[22], virtual laboratories may require improvement in this aspect to work effectively.

Additionally, virtual laboratories are intrinsically unable to provide students with the real-world hands-on experiences with laboratory equipment and practices that they would later be required to use, such as correct pipetting and microscopy techniques. The importance of acquisition of these skills is debatable however, as while some studies argue that the 'the absence of wet lab experiences also limited students' ability to grasp conceptual knowledge'[21], others have found that 'the focus of the labs is on acquiring technical and diagnostic skills, and the cognitive load imposed by learning to use real lab apparatus is possibly redundant to that goal.'[23] The importance of

acquisition of practical skills will not be explored further in this paper, as it is not an aspect of practical work that virtual laboratories can assist in teaching regardless.

1.5 The aims of this study

In this project, a series of virtual laboratories will be created specifically with active learning and meaningful consequence as its primary tenets, and their efficacy will be investigated. This will be done by creating multiple virtual laboratories, each tailored to a specific current undergraduate level online university course. The virtual laboratories will be created with permission and cooperation from course coordinators and aim to allow the students using them to have complex interactions with the virtual laboratory, experiencing consequence and learning through repeated cycles of prediction, action, feedback and reflection.

After completing their experiments online within the virtual laboratory, students will be asked to fill out a feedback form that examines how they perceived they interacted with the virtual laboratory. The questions will investigate how long their interactions with the virtual laboratory lasted, whether they were able to iteratively improve their output from the virtual laboratory by analysing their obtained results and if they would feel more confident conducting a real version of their experiment after using the virtual laboratory.

The respective academic course coordinators will also be asked to fill out extended responses regarding the challenges they faced in delivering online laboratory classes without the virtual laboratory, and how well they found the virtual laboratory to address these challenges.

2 Materials and Methods

The process of creating each of the virtual laboratory tools that were used in the study followed the ADDIE approach, for *Analyse, Design, Develop, Implement,* and *Evaluate*. This method is applicable to the development of learning resources, and is specifically used to help guide educational staff in the development of effective learning environments[1]. Each step in the process is used to guide the next, as detailed in Figure 2-1.

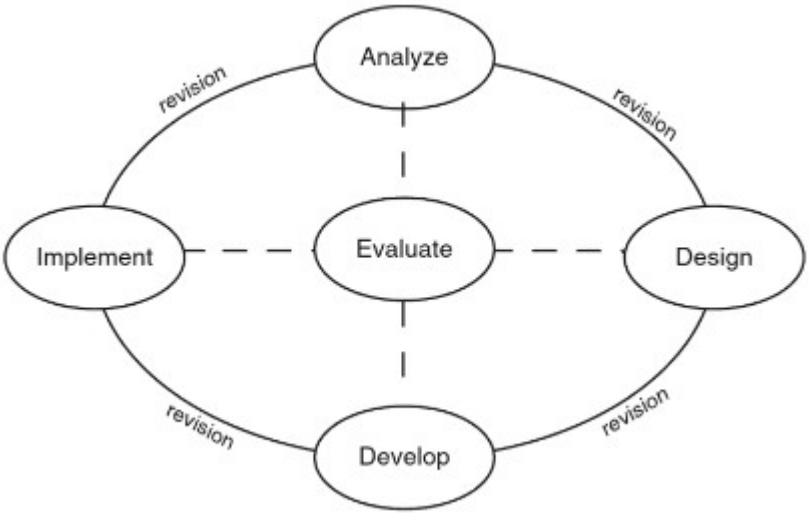


Figure Fehler: Verweis nicht gefunden-1 | The ADDIE Concept[1]. The ADDIE cycle is a framework that has been applied to systematic product development and is an effective process for the development of learning resources such as the virtual laboratories being designed in this project.

2.1 Context of the study

The study was conducted within three separate third year undergraduate biochemistry courses. In each of these courses, a virtual laboratory tool created to be used by students as a substitute for a physical laboratory. In previous years, each of the courses had run these laboratories in face-to-face environments for students but were unable to do so in 2021 due to social distancing requirements imposed in response to COVID-19.

2.1.1 BCMB3X01 Gene and Genome Regulation

The first course in which a virtual laboratory was used was titled 'BCMB3X01 – Gene and Genome Regulation.' This course was targeted towards fifth semester undergraduate Bachelor of Science Students and was composed of both a mainstream and advanced component. The course aimed to provide an in-depth look into the central dogma of molecular biology, investigating the different genetic mechanisms by which protein expression is ultimately controlled. The course was accompanied by a practical component, with four laboratories taking place throughout the semester, each related to these learning outcomes.

The final laboratory session in this course included a western blot experiment, and this was the practical targeted to be replaced by a virtual laboratory for this project. The experiment investigated the changes in expression levels of target proteins Aminolevulinate synthase 2 (ALAS2), ß-globin, Carbonic Anhydrase 1 (CA-1), Nuclear erythroid factor 2 (NFE2), and Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) in mouse erythroleukemia (MEL) cells as they progressed through dimethylsulfoxide (DMSO) induced erythropoiesis.

In previous years when the laboratory was conducted face-to-face, students were provided with four MEL cell samples that were either untreated or DMSO (1.8% v/v) treated for 24, 48, or 72 hours and in the first steps, they performed a simple protein extraction. Next, they quantified the amount of protein in each sample using a BCA protein assay and separated the proteins via an SDS-PAGE gel. After this they would transfer the proteins to a membrane, block the membrane with a 5% milk solution and then probe the membrane with antibodies specific to each of the target proteins. HRP-conjugated secondary antibodies were then added to the membrane, followed by HRP substrate. The blot would then be imaged, and the images saved in the student's electronic laboratory notebooks. These images could then be quantitively analysed visually compared to determine the relative quantity of each protein at each timepoint, and these results discussed.

2.1.2 BCMB 3X03 Biochemistry of Human Disease

The second course targeted for virtual laboratory development was 'BCMB3X03 – Biochemistry of Human Disease' and also consisted of both a main and advanced stream of students. In this course, the underlying cellular imbalances behind human diseases were investigated, with a focus on how the malfunction of different cellular functions can cause various diseases. The course included a practical component of three laboratories that explored these concepts. The laboratory replaced in 2021 by the virtual laboratory was a western blot experiment. In this experiment, the effects on cell survival of resveratrol, prednisolone, and a combined treatment of resveratrol plus prednisolone on viability of a human acute lymphoblastic leukemia (ALL) cell line were investigated. In the laboratory, students were supplied with ALL cells and tasked with performing one of the 8 different treatments including two concentrations of each resveratrol or prednisolone treatment, two concentrations of combined treatments, or a media or DMSO control. These treatments were each loaded onto an SDS-PAGE gel by students, and subsequently transferred to a blot membrane. Membranes were probed with rabbit anti-Caspase 3 and rabbit anti-alpha tubulin antibodies, followed by peroxidase conjugated anti-rabbit antibodies and ECL substrate. After the substrate was applied, colour formation revealed bands corresponding to Caspase 3(32 kDa), Cleaved Caspase 3 (10kDa), and alpha tubulin (50 kDa). The cleavage of Caspase 3 is an indicator of apoptosis that allowed students to determine the relative efficacy of each treatment, while alpha tubulin acted as a loading control for the samples. Students would then save a copy of the imaged blot into their electronic notebook to annotate, analyse and discuss.

2.1.3 BMSC 339 Cellular Regulation

The last course, 'BMSC339 – Cellular Regulation' focused on molecular processes responsible for upkeep of cellular function, and how abnormalities could lead to disease. Proteomics experiments were a core part of the course, which investigated how they could be used as an indicator of different cellular functions. In this course, the virtual laboratory was used in place of a two-part immunochemistry laboratory, that consisted of both immunoprecipitation and immunodetection protocols. In the two-part protein immunochemistry laboratory, the affinity of anti-bovine albumin

antibodies for bovine, pig and chicken albumin was investigated. Part A was an immunoprecipitation experiment, aiming to confirm the presence of albumin in a bovine serum sample. The bovine serum was centrifuged, and anti-bovine albumin antibodies were then applied to both the supernatant and pelleted components. These antibodies were then bound to Protein A conjugated to polymer beads, which were then separated and stained after SDS-PAGE and analysed for the presence of bands corresponding to albumin.

2.2 Analysis of candidate course requirements

When beginning to design and outline the virtual laboratories it was crucial to first define the specific learning objectives of each course and laboratory. As the learning outcomes of the practicals varied between the courses, it was important to communicate with coordinators throughout this process to define these goals. This ensured that each of the laboratories focused on the content relevant to the course within which it ran and would help students acquire the theory of practical knowledge and to maintain established links to theoretical aspects of the course.

~~2.2.1~~

2.2.2 BCMB3X01 Gene and Genome Regulation

As BCMB3X01 was the first participating course, the process of defining the scope of the virtual protocol presented various challenges. Critically, we needed to identify the reason for inclusion of the practical class, what students were expected to learn upon completion of the activities and to identify any established links to the lecture component of the course. The first step of the process was to collate the written learning outcomes relevant to the practical. The BCMB3X01 course outcomes that pertained specifically to the practical component were:

- To collect experimental data and adapt, develop and trouble-shoot experimental procedures for novel contexts and requirements.
- To assess the quality of data, interpret and draw conclusions from data obtained in the laboratory.

In the outcomes of this course, the primary focus is on analysing data obtained by students themselves in the laboratory. This is consistent with the previously defined

goal of creating a virtual laboratory that would allow students to generate their own data that they could analyse, and as such would be focused on during the development of the virtual laboratory. The experimental decisions that they would make during the protocol would also have to affect the data generated, to allow students to meaningfully exercise and test their theoretical knowledge as outlined above.

2.2.3 BCMB3X03 Biochemistry of Human Disease

The western blot practical in this course had similar outcomes to that of BCMB3X01, though the context of the investigation varied between the courses, leading to different requirements for students in this cohort. The course outcomes relating to the practical are as follows.

- To perform and evaluate a biochemical assay to determine protein concentrations in solutions using different approaches.
- To evaluate and communicate experimental data by composing a scientific paper based on your own experimental results.
- To critically evaluate experimental data.

This learning activity also aimed to give students an opportunity to reflect on their own experimental data, with the evaluation stage being the primary focus. As such, a virtual laboratory that provided students with an output that they were able to analyse was the primary focus for the virtual laboratory development.

2.2.4 BMSC339 Cellular Regulation

In the Cellular Regulation course, the practical learning outcomes were

- Demonstrate practical skills relevant to cell biology and be able to act safely in a laboratory environment.
- Describe the principles of signal transduction in cells and use them to explain the operation of selected signalling pathways.
- Design and interpret experiments to investigate cell signalling.
- Interpret events, facts, and experimental data relevant to the above topics.

The outcomes of this course included those of acquiring and analysing data outputs, with an additional aim toward the development of student's practical skills. This would be considered when moving further through the design stages of this virtual laboratory.

Overall, the practical learning objectives that were used in designing each of the virtual laboratories were similar between their respective courses. Each course aimed to expand student's knowledge through the process of designing and implementing a protocol and acquiring an output set of data that could be used to analyse both the aim of specific laboratory and the success of their experimental design and methodology.

2.3 Designing an effective virtual laboratory

Once the goals of each course were identified, the scope of each of the virtual laboratories was then defined. This involved deciding where in each protocol the virtual laboratory would start and finish, the steps of the protocol that students would be able to manipulate, and what effects these decisions the students made would have on subsequent outputs.

2.3.1 BCMB3X01 Gene and Genome Regulation

To decide on the specific experimental context in which these broader learning goals would be achieved, the additional learning outcomes relating specifically to this laboratory were considered. These were 'Studying a key biochemical and molecular response of MEL (Mouse Erythroleukemia) cells to stimulation by DMSO (dimethylsulfoxide),' 'The technique of Western Blotting to analyse proteins in cell lysates' and 'observing the changes in the proteome of a cultured cell line.'

Through consideration and discussion with the course coordinators on how to best achieve learning outcomes, it was decided that the focus of this virtual western blot laboratory would be on providing students with a virtual substitute for the protein quantification, gel loading, and blotting procedures. These specific steps were elected to be the three main constituent components of the virtual laboratory, as they were

the areas of the protocol in which the students had to make the most consequential choices based upon their theoretical understanding of the biochemical processes. They would additionally allow students to meet the specific learning outcomes of the laboratory, by engaging with the practical techniques while also considering the underlying biochemical principles.

The initial protein quantifications step aimed to provide students with as similar an experience to the physical laboratory as possible, as quantifying protein samples with a self-constructed absorbance standard curve had been a staple of the physical laboratory in previous years. This would be done by providing students with four protein samples at unknown concentrations from each timepoint, a BSA protein sample at a known concentration, and a buffer solution. Students would then be able to dilute these samples as desired and measure the resulting absorbances. They would then be able to use the absorbances from the known concentration sample to calculate the concentrations of each of the samples from the four timepoints.

In the subsequent loading stage students would be presented with the same MEL samples from each timepoints, samples from each of the proteins of interest (Alpha globin, ALAS2, NFE2, CA1, GAPDH) at known concentrations, a standard ladder solution, a loading buffer and RIPA buffer. Using these solutions and a set of test tubes, students would be able to dilute and mix these solutions as they saw fit, before loading a desired amount into each lane of the SDS-PAGE gel. The goal would be for students to load each of the samples at an appropriate concentration with loading buffer, so that the gel could be later used for immunodetection. It would be crucial here to track the volumes and concentrations of each solution that students moved between vessels, so their outputs in future stages would be as authentic as possible. Once the gel was loaded, they would specify the time for and voltage at which to run the gel, which would then take them to the next stage displaying the gel output.

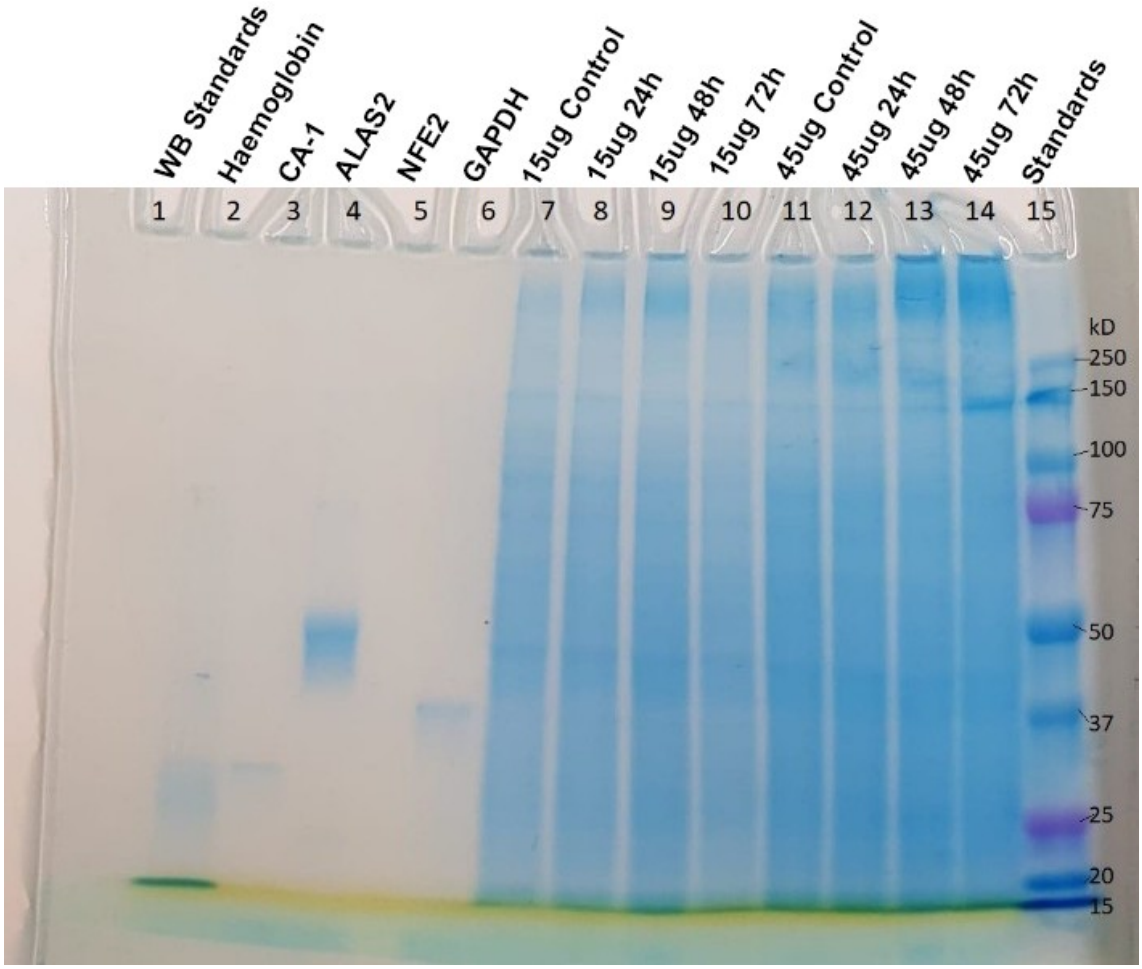


Figure Fehler: Verweis nicht gefunden-2 | BCMB3X01 SDS-PAGE Gel. Gel image from a previous class, showing isolated protein control bands in the first five lanes and whole protein sample smears in lanes 7-14.

In this stage, the students would be presented with their stained gel image from the last step, with visible bands of protein appearing if the previous step was completed correctly. If the gel had been loaded incorrectly then the output would have to represent this, for example if no protein was loaded into one of the lanes by accident, then the lane would appear empty. Students would then have the choice to proceed to the blot stage with this gel or return to the previous stage to reattempt the loading process.

After confirming that they would proceed with a gel, they would then need to transfer the gel to a membrane before the blotting process. This would test students' knowledge on the order of the transfer cassette, though how this would be represented graphically was not yet decided.

Once the transfer was completed, the next and final stage would be that of the blotting process. Here students would be able to interact with their membrane in different ways, with the goal to acquire an imaged blot with their targeted proteins visible. This would ideally be done by blocking the membrane, applying the correct primary and secondary antibodies at the correct concentrations, applying the HRP substrate of their secondary antibody and incubating, and then imaging the membrane, with washing steps in between. These options would all be available to the student through buttons they could press, with input fields to specify the time or concentration of the process when applicable. Mis ordering any of these steps could potentially lead to acquiring a blank blot image, and so this would require students to understand the protocol and test various orders and antibody concentrations. It was additionally decided to withhold the optimal concentration of each antibody, allowing students to optimise these concentrations on their own by repeated attempts. When the process was successfully completed, students could analyse the results and comment on the quantity of protein present in each sample and why this might be. The outputs could also be screenshot and placed in the student's electronic notebook, with teaching staff using this to assess their success in the laboratory.

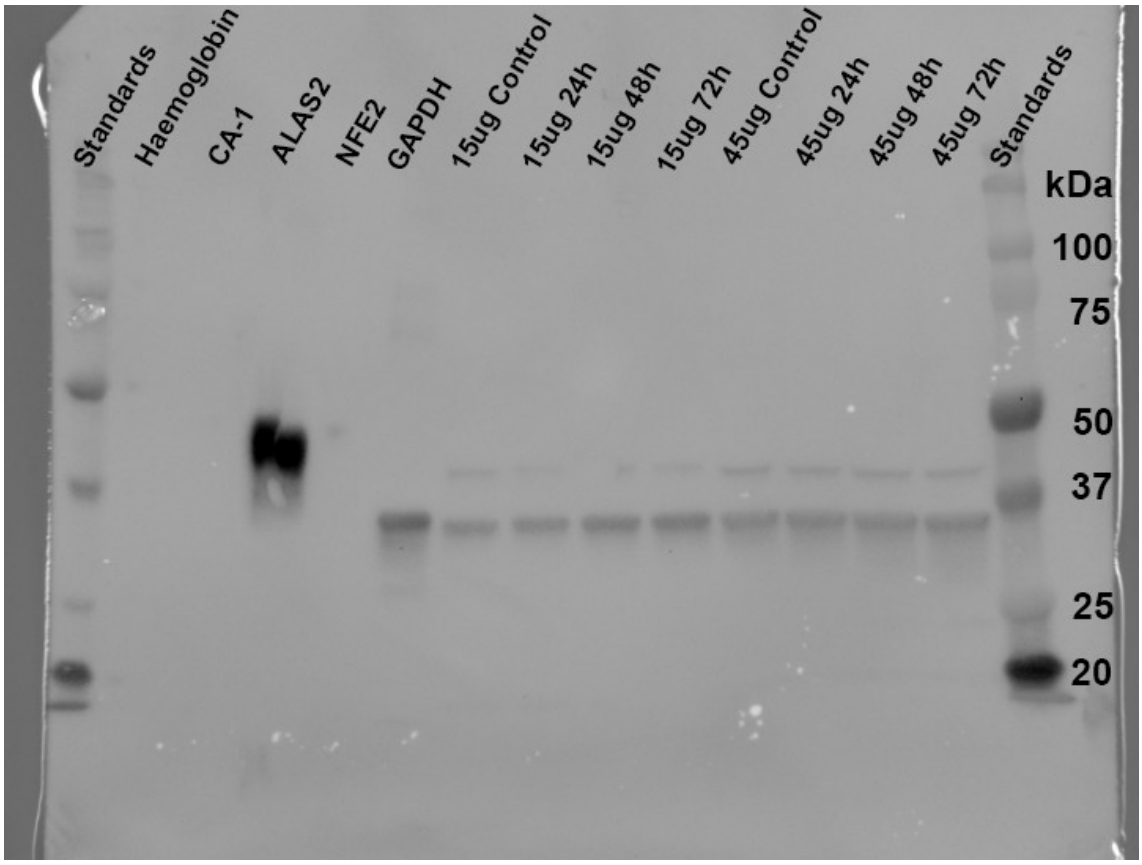


Figure Fehler: Verweis nicht gefunden3 | BCMB3X01 Imaged Blot. A blot from a physical BCMB3X01 class in a previous year, probed for ALAS2. Note dense ALAS2 protein sample in lane 4, and bands corresponding to smaller ALAS2 subunit that was present in the samples.

2.3.2 BCMB3X03 Biochemistry of Human Disease

The virtual laboratory for BCMB3X03 was originally intended to consist of two sections of the larger practical, a flow cytometry protocol and an immunoblot. Their respective learning goals are 'the use of flow cytometry to determine cell viability and apoptosis' and 'the use of Western blot/immunoblot to determine protein levels and protein activation.' Both of these practicals take place in the context of the combinatorial treatment study of prednisolone and resveratrol on cell viability, with the flow cytometry used to determine cell survival after treatment, and the western blot used to detect the presence of cleaved caspase-3 and alpha tubulin as indicators of cell death.

During development of the previous laboratory, there were difficulties in defining the appearance of the virtual laboratory, and which tools the students would be presented with at each screen. To aid in the development of the BCMB3X01 virtual laboratory, a mock-up overview was created using PowerPoint prior to development beginning, to

ensure all required aspects of the laboratory were included, and act as a scaffold during development.

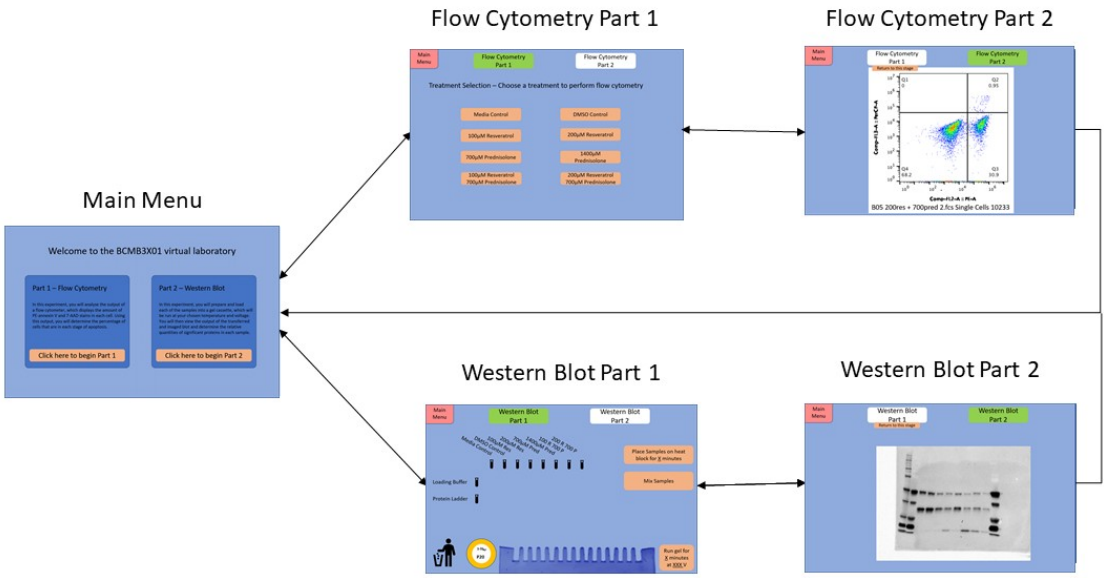


Figure Fehler: Verweis nicht gefunden-4 | BCMB3X03 Program map. Mock-up program outline of the BCMB3X01 laboratory, showing user navigations between parts and rough overviews of each scene. The Flow Cytometry component was later scrapped due to time constraints.

As shown in Fig X, virtual laboratory would consist of both the Western blot immunodetection stage and flow cytometry stage, which could both be accessed from a main menu screen.

The Western blot part was similar to the one used for BXMB3X01, though did not begin with a quantifications step. The first stage would consist of both sample preparation and gel loading. Students would be presented with the 8 different Resveratrol/Prednisolone combined treatments, in addition to a standard ladder and a loading buffer. These treatments would need to be prepared by adding a fixed amount of loading buffer to each and setting them on a heat block to allow the proteins to unfold. They could then be stored in an ice bath to hold the proteins them in this state before they were loaded onto the gel. If the proteins were not allowed to heat for a sufficient time on the heating block, or allowed to cool on the bench, the results of the gel would become smeared as in reality. Once the gel was loaded, a button would

allow students to run it for 45 minutes at 175V, upon which they would be taken to the next stage.

The second stage was originally intended to take students directly to the blot output, as shown in figure x. This stage was later expanded into three individual stages of a gel output, a transfer stage, and a blot stage with all blotting options to be used by students as in BCMB3X01 detailed in 6.3.1. In this version, however, the correct concentrations of antibodies were applied automatically, as this was not important within the context of the course. Once students had acquired a successful blot output image, this could then be screenshot and placed within their electronic notebook for assessment.

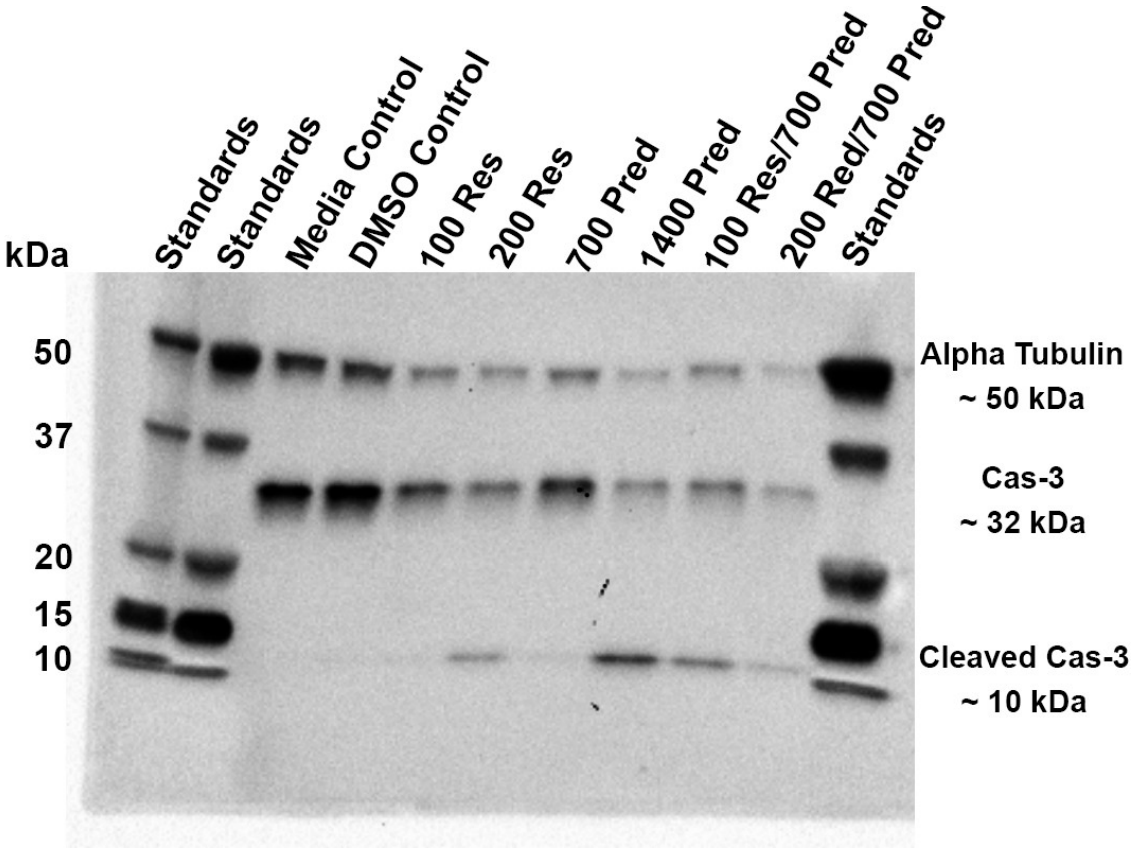


Figure Fehler: Verweis nicht gefunden-5 | BCMB3X03 Imaged Blot. An imaged blot from a previous practical class, probed with anti Cas-3 and anti-Alpha tubulin antibodies. Note increased Cas-3 cleavage in 1400 prednisolone and both combined treatments.

2.3.3 BMSC339 Cellular Regulation

The BMSC339 laboratory did not have specific learning outcomes within the immunochemistry laboratory, and so the outlining of the online laboratory was done

through communication with teaching staff. As the identified requirements for this virtual laboratory had an additional focus on laboraty techniques and sample preparation, it was decided that the virtual laboratory would need to allow students to complete a more in depth preparations step than in previous virtual laboratorys.This laboratory would cover both the immunoprecipitation and immunoblotting stages of the laboratory, with sample preparation stages for each, as seen in Fig x.

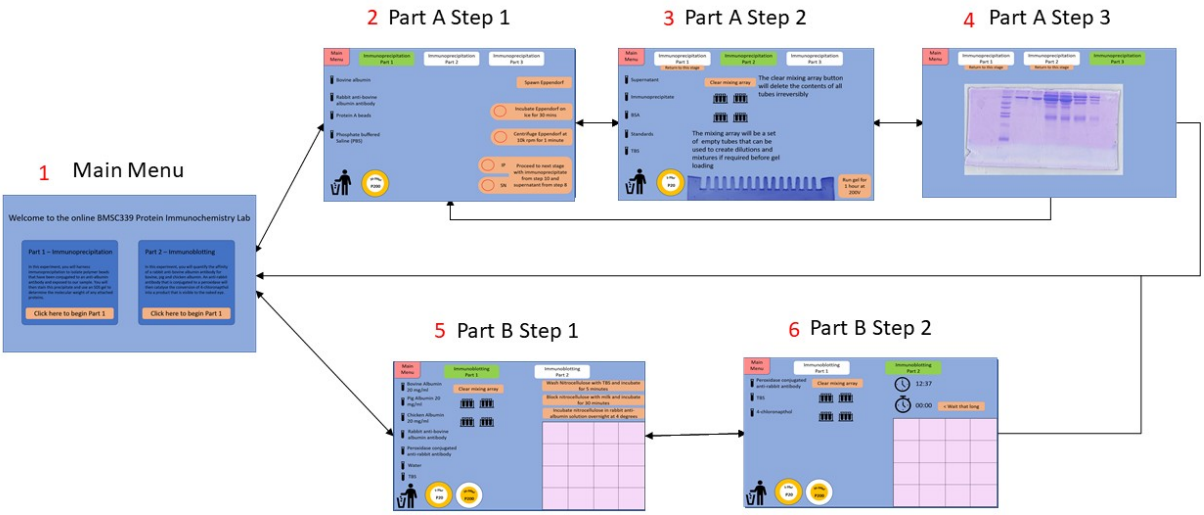


Figure Fehler: Verweis nicht gefunden-6 | BMSC339 Program map. Mock-up overview for BMSC339 Immunochemistry laboratory.

In the immunoprecipitations part of the laboratory, the aim was to determine if albumin was present in a supplied bovine serum sample. The first stage involved preparing the bovine serum sample to be run on an SDS-PAGE gel and involved separating the supernatant and pelleted material via centrifuging and binding potential albumin with anti-albumin antibodies and protein A beads. Students were supplied with each of these three solutions and phosphate buffered saline (PBS) to act as a buffer solution, with interface buttons allowing them to incubate specific Eppendorf tubes on ice for 30 minutes or centrifuge them as required. Through following their designated laboratory protocol, they would finish this stage with two Eppendorf tubes corresponding to the supernatant and resuspended pelleted material from the bovine serum sample. These two resulting Eppendorf tubes could then be brought into the subsequent loading stage, so correctly executing the preparation in this stage would be vital to successfully completing this section of the practical.

In the gel loading stage, students would have access to their supernatant and pelleted samples, in addition to whole serum samples, a standard ladder, and a TBS solution. They would also have pipettes and mixing tubes at their disposal to mix and dilute these solutions before loading as they saw fit. The laboratory protocol called for the supernatant, pelleted material and serum samples to each be loaded into individual wells as quantities of 5uL and 20uL with a single lane of standards, though students were able to load the lanes as they saw fit. Once the student was satisfied with what had been loaded into the gel, they would then be able to press a button that ran the gel for one hour at 225V and stained it with Coomassie blue, presenting them with the output.

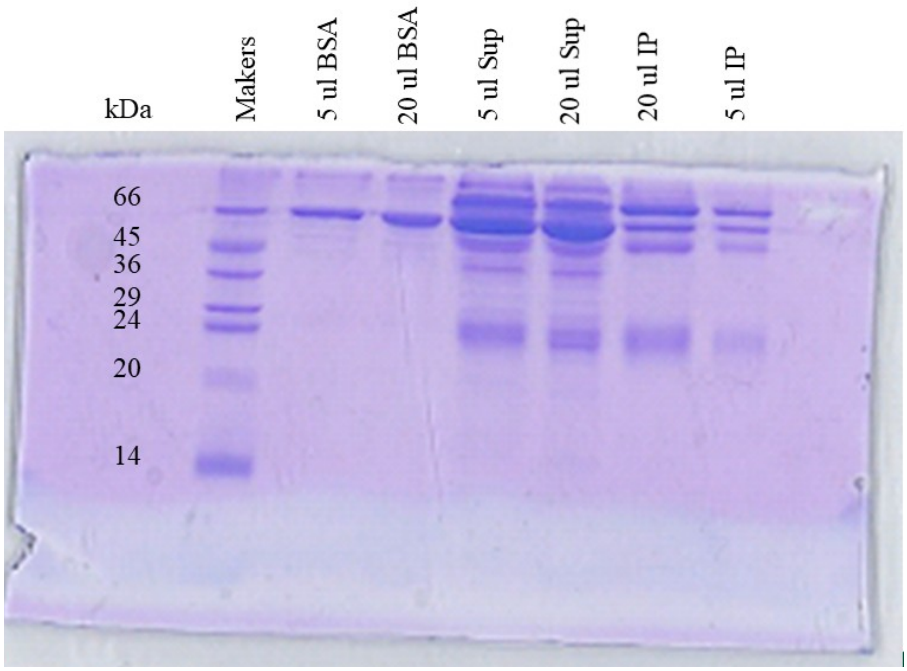


Figure Fehler: Verweis nicht gefunden-7 | BMSC339 SDS-PAGE gel. SDS-PAGE gel result from a practical class in a previous year.

An example of a stained gel output from the laboratory in previous years that was loaded as per the protocol can be seen in figx, though it would be important for student to receive a gel corresponding to their methodology in the sample preparation and loading stages. It is important to note that not all deviations from the protocol would be detrimental to students completing the task, and experimentation was encouraged to allow students to test their ideas. Students would then be able to place

this gel output into the electronic notebook and comment on the presence of albumin in the samples.

The Immunoblotting section of the laboratory involved the use of a dot-blot protocol to probe bovine albumin, pig albumin, and chicken egg albumin with anti-bovine-albumin antibodies. The results would be used to assess and comment on the binding affinity of the anti-bovine-albumin antibodies for each of these homologous albumin proteins. In the first stage of this protocol, students would have access to each of the three albumin species, both primary and secondary antibodies, water, TBS, and a 10% milk powder solution in TBS. There would also be a clock with buttons allowing the flow of time to be sped up and slowed down, and interface buttons allowing students to pour solutions off the membrane or incubate the membrane overnight that would take students to the next scene. The protocol began by the spotting of undiluted, 1/10 and 1/100 dilutions of each of the three proteins and controls of TBS and both primary and secondary antibodies onto a nitrocellulose membrane. After allowing the membrane to dry, it would then be blocked with the 10% milk solution for 30 minutes before washing off to prevent unwanted binding to the membrane. The anti-albumin primary antibody would then be applied to the membrane and allowed to bind overnight, upon which also proceeded students to the final stage.

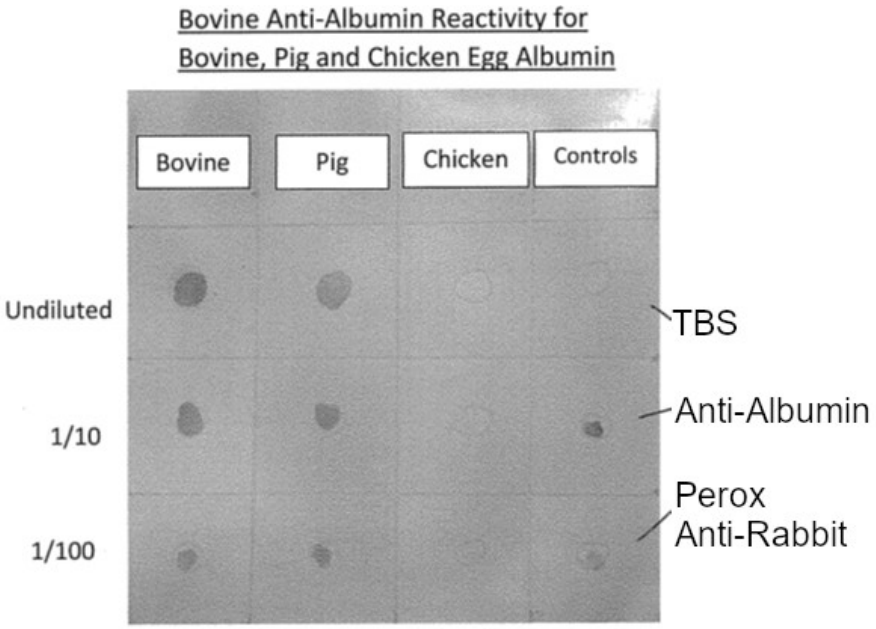


Figure Fehler: Verweis nicht gefunden-8 | BMSC339 Dot Blot result. A BMSC339 dot blot result from a previous year probed with anti-bovine-albumin antibody. Dots are darker in the bovine column than pig at all dilutions, while chicken does not appear at any dilution level.

The goal of the final stage would be to apply the secondary antibody followed by it's substrate to cause the appearance of visible dots to appear at the locations at which the primary antibody had bound, as seen in figure x. The students would again be provided with the required solutions and the virtual laboratories to dilute and apply them to the membrane, in addition to a clock to monitor the current time and change the speed at which it passed. This step would be done by first washing away the primary antibody solution that had been on the membrane overnight, then applying a solution of TBS containing the anti-rabbit peroxidase conjugated secondary antibody to the membrane for 30 minutes. Once this was also discarded and the membrane washed, the membrane would then be incubated in a diluted 4-chloronapthol solution until the catalysation of the 4-chloronapthol into visible coloured precipitate could be seen. This step could take up to an hour to pass depending on the quantity and dilution factor of all reagents used in previous steps, and would be gradual, with transparent dots growing opaquer over time. Once the dots had appeared fully students could screenshot this and obtain results like those seen in figure x, if the protocol had been followed correctly.

2.4 Virtual laboratory development

Once these outlines of each virtual laboratory had been completed, the development stage began in Unity. Unity is a game development engine and was chosen as the backend for these virtual laboratories due to its capabilities as a 3d engine that could represent a benchtop laboratory space well, and the ability to export the virtual laboratory in a format that could be hosted on an online server that students could access remotely. Development took approximately six weeks for each of the virtual laboratories from the end of the design stage to the viable product, though this decreased slightly for each subsequent instance. In this methods section some of the primary challenges are detailed in the scripting of each of the virtual laboratories, though this list is not exhaustive, and many smaller challenges were also dealt with. Additionally, many elements created for the first virtual laboratory were recycled to be used again in those following, and as such each feature is detailed within its original environment.

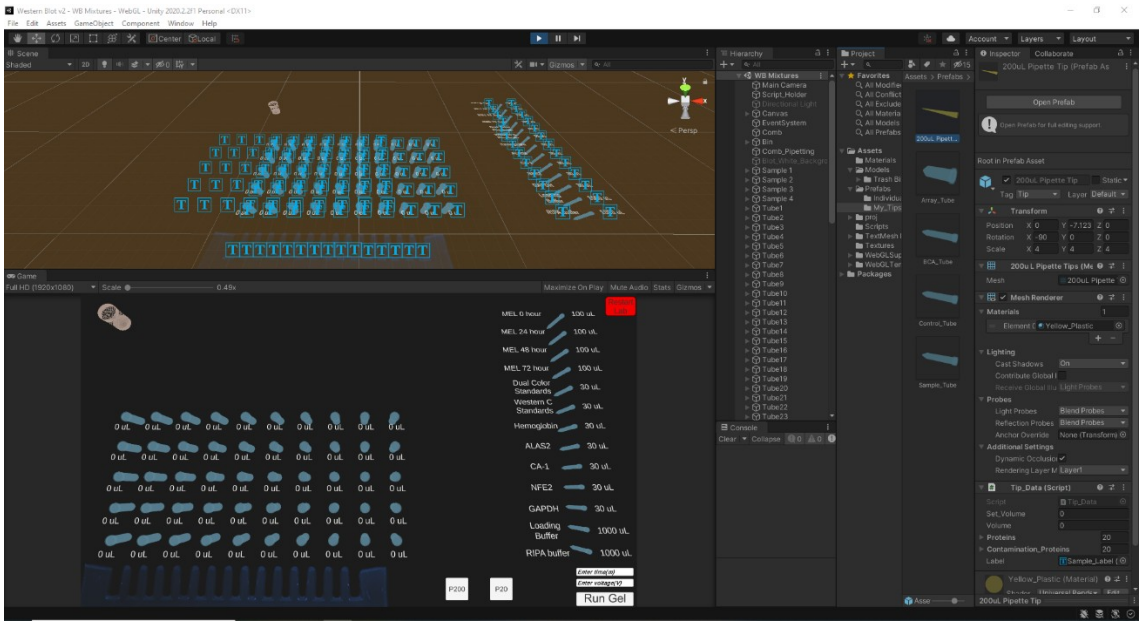


Figure Fehler: Verweis nicht gefunden-9 | View within Unity program. An example of the development process for each of the virtual laboratories. The user view can be seen in the bottom left, with the scene view above.

2.4.1 BCMB3X01

2.4.1.1 Creating a gel output

The first step in creating a virtual laboratory for BCMB3X01 was ensuring that a viable SDS-PAGE gel output image could be generated, akin to the example seen below. The gel output would not be labelled like the example, as that would be up to the students to attempt themselves.

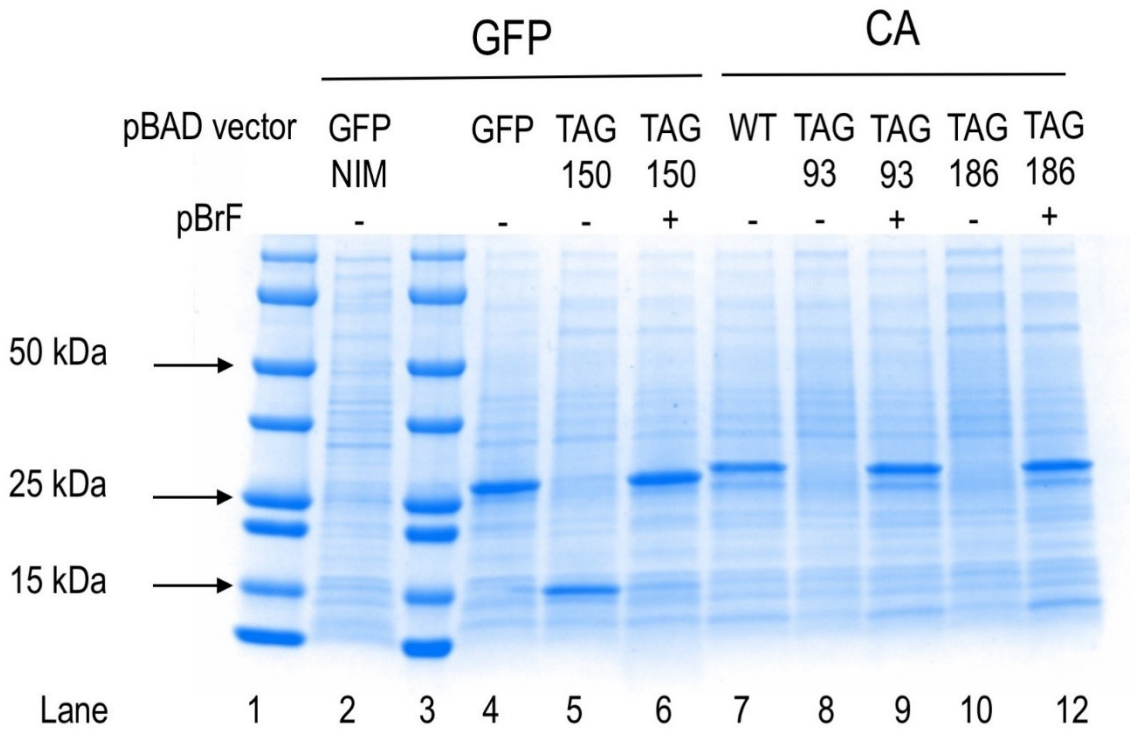


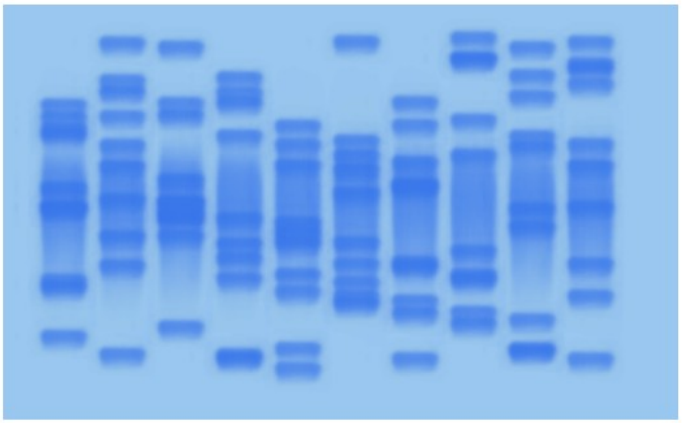
Figure Fehler: Verweis nicht gefunden-10 | Example of physical gel. An physical stained SDS-PAGE gel output[24].

The most basic element of a stained SDS-PAGE gel is that of the protein band, being the fundamental unit that comprises a gel image. Each band in an SDS-PAGE gel represents one or more proteins from the sample that have migrated the same distance through the medium. A single lane in a gel can be made up of any number of individual protein bands of various levels of visibility, depending on the sample loaded. A gel is typically comprised of 6 – 15 lanes of protein, each aligned horizontally as in Figure 2-10.

To begin the process of virtually synthesising a passable output image began with the isolation of a single band image. This single band could be replicated as many times as necessary at random heights in a vertical column to produce what appeared to be a single lane of proteins. Repeating this process multiple times horizontally to create lanes would result in a basic virtual gel image, the first output of which can be seen in fig x

A

B



C

```
public void Make_Band()
{
    GameObject ProteinBand = Instantiate(Band_Background_Prefab,

    new Vector3(FirstTrackX + DistanceBetweenTracks * (Track - 1), UnityEngine.Random.Range(1,8) , 0),

    Band_Background_Prefab.transform.rotation);

}
```

Figure Fehler: Verweis nicht gefunden-11 | SDS-PAGE gel output simulation. a, Isolated image of a single band. **b,** Virtual SDS-PAGE gel image created through instantiation of many randomised band images. **c,** Code responsible for spawning bands, with x value based on 'Track' integer and y value randomly generated between 1 and 8.

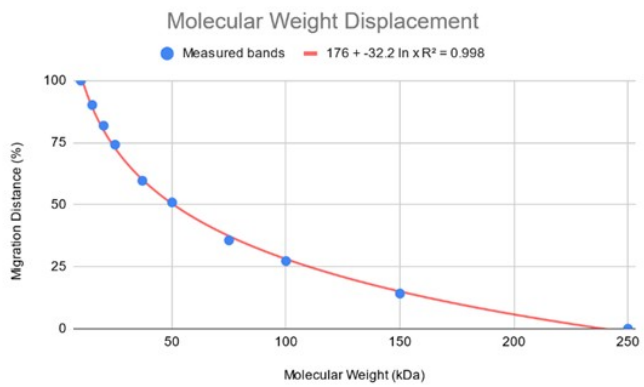
After randomised gels could be generated, the next step was to control the height at which each band spawned, to allow the creation of bands representative of the specific proteins that had been loaded into the gel. In Figure 2-11 B, the height of each band is randomly generated at between 1 and 8 y. To control the height at which each band was spawned, a function needed to be created that took in a molecular weight and output a corresponding height value between 1 and 8. This was done by measuring the migration of a standard ladder of known molecular weights from a real gel photo and converting the migration to a percentage of the farthest distance

migrated. The data was then graphed, and the logarithmic function that best fit this migration found.

A

kDa	% dist from top	Pixels from top
250	0	0
150	14.2	39
100	27.3	75
75	35.6	98
50	50.9	140
37	59.6	164
25	74.2	204
20	81.8	225
15	90.2	248
10	100	275

B



C

```
public float kDa_To_Y(float kDa)
{
    return (BandMaxHeight - ((BandMaxHeight - BandMinHeight) * (176 - 32.2f * (float)Math.Log(kDa)) / 100));
}
```

Figure Fehler: Verweis nicht gefunden-12 | Conversion of molecular weight to height. a, Standard ladder measurements. The middle column is derived from the pixels measured to the right. **b,** Graphed measurements in blue, accompanied by trendline of best fit in red. **c,** The code responsible for the conversion of kDa to the corresponding height.

A new function was then created that used the function to convert the molecular weight of a protein in kDa to a percentage of the total migration (Figure 2-12). This meant that when proteins were loaded into a well that had molecular weights assigned to them, the corresponding bands would appear at the correct height. Additionally, the opacity of each of the bands was made to be proportionate to the quantity of protein that was loaded into the gel, allowing smaller amounts of protein to appear fainter on the gel.

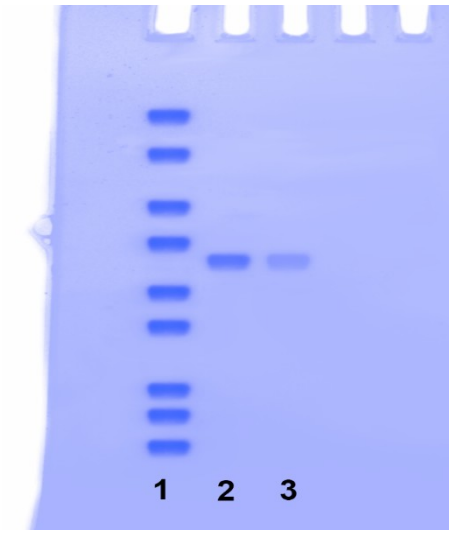


Figure Fehler: Verweis nicht gefunden-13 | Virtual Standard ladder and albumin. Virtual blot output of a standard ladder in lane 1, with 15µg and 5µg of albumin in lanes 2 and 3 respectively.

The control of both the height and opacity of bands now allowed gels to be generated faithfully according to the proteins in the solutions that were loaded into each gel (Figure 2-13).

2.4.1.2 Pipetting fluid between vessels

The next step was to give the user the ability to move fluids correctly between different tubes, so that solutions could be mixed, diluted, or moved as necessary. This was done by creating an equation that was called each time a solution was pipetted out of or into the container. When pipetting down into a container, the volume of the pipette and quantities were added to those already in the container, and the pipette emptied. When pipetting upward from a container, the equation divided the volume the pipette was set to by the volume in the container and multiplied the quantity of each protein in the container by this amount. For example, if 10 µL was being pipetted out of a container containing 5 µg of protein diluted in 25 µL of total fluid, then 10 / 25 * 5 µg = 2 µg of protein were moved into the pipette tip. This amount was then removed from the container to stop generation of additional fluid.

2.4.1.3 BCA Assay calculation

One of the goals of this virtual laboratory was to require students to quantify their protein samples through the use of a BCA assay. This is done by measuring the

absorbance values of a protein of known quantity, calculating the relationship between them, and then reversing this equation to obtain the unknown protein quantities from their absorbances. For students to be able to do this, an equation was written that simulated a spectrophotometer by converting a quantity of protein to an absorbance value. The equation obtained used was ABS = 0.00424 * X µg/mL protein, meaning that 1mg/mL of protein would yield a theoretical absorbance of 4.24. As this absorbance spectrophotometer used in the class capped at an absorbance of 1.9, the equation was modified to reflect this, and values above 1.9 would return 1.9 nonetheless. Additionally, a random number between -0.011 and 0.011 was added to the final absorbance, providing an error margin equal to the machine in the dataset used.

This allowed the user to dilute their samples and standards using the pipette as desired, and obtain corresponding results. A serial dilution of 1000 µg/ml to 31.25 µg/ml was recommended, though any dilution range below the maximum absorbance of 1.9 allowed users to construct their standard curve using the data.

2.4.1.4 Blotting options

The final stage of the BCMB3X01 laboratory was the western blot, where students would have many different ways to interact with their proteins that had been transferred from the SDS-PAGE gel to a membrane. These options included blocking the membrane with milk, washing the membrane, applying primary or secondary antibodies to the membrane, incubating the membrane with HRP substrate, or exposing the membrane for imaging. Multiple of these options involved additional parameters that had to be able to be chosen by students, including the choice of antibody to apply, the ratio at which the antibody was dilution before applied, and the time for which the membrane was incubated, washed, or imaged for.

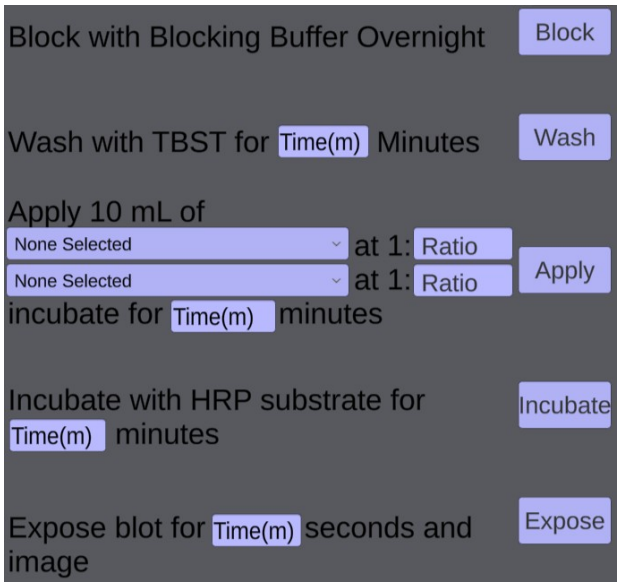


Figure Fehler: Verweis nicht gefunden-14 | Blotting options interface. The control panel available to users through which the blotting process was done. The time and ratio field were able to be clicked and typed into, while the antibody selection was a dropdown list that allowed selection from the following antibodies: Mouse anti-beta globin, Mouse anti-CA1, Rabbit anti-ALAS2, Rabbit anti-NFE2, Mouse anti-GAPDH, Goat anti-mouse IgG, HRP-conjugated, Goat anti-rabbit IgG, HRP-conjugated.

Whenever a user would execute one of the methods via the buttons on the right in Figure 2-14, the corresponding process would be applied to the gel if applicable. When the gel was incubated in an antibody, the gel was searched for the presence of corresponding binding partners and the antibody attached to them if they were found. If there were no suitable binding partners, such as in the case where the secondary goat anti-mouse antibody was applied before any mouse antibodies, then nothing would happen. The quantity of antibody bound was also influenced by both the time for and ratio at which the membrane was incubated, and adjusted correspondingly, meaning shorter or more dilute incubations led to less binding. If both primary and secondary antibodies had been applied to the membrane followed by incubation in HRP substrate, then the membrane could be successfully imaged (Figure 2-15). When exposed, the black bands would appear on the blot where the HRP-conjugated secondary antibodies had produced imageable product.

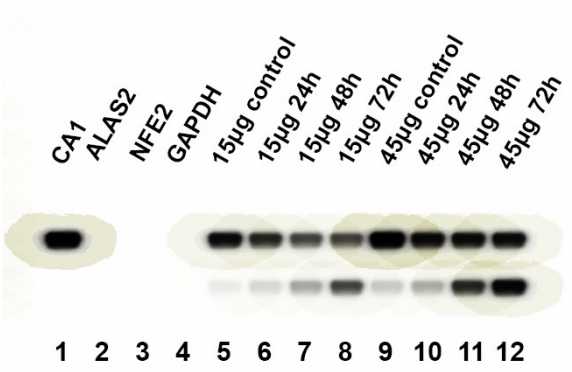


Figure Fehler: Verweis nicht gefunden-15 | Imaged blot. A successfully loaded and imaged virtual blot that had been probed for beta globin and CA1. The upper visible bands correspond to CA1(30 kDa), while the lower bands are beta globin(16 kDa). Note the decrease of beta globin over the course of each treatment, an important laboratory outcome.

2.4.2 BCMB3X03

2.4.2.1 Unfolding proteins using a heating block

In this virtual laboratory the transfer and blotting stages were similar to those used in BCMB3X01, whereas the sample preparation stage required new functionality. In the preparation stage, students were required to unfold their proteins by placing them on a heating block for roughly five minutes, and then cool them quickly in an ice bath, to capture the proteins in this unfolded state. To allow students to control how long the samples were placed on the heating block in the program, a clock was added to the interface of the program, allowing time to be sped up or slowed down as required (Figure 2-16). Once the clock was created, a script was written that affected each sample within the heating block. This script would convert folded proteins to unfolded proteins at a rate of 30% per minute while they were in the heating block. As this function was compounding, after five minutes roughly 85% of the protein would be converted to an unfolded state. If these samples were removed from the heating block and placed onto the benchtop, this reaction would begin to reverse at the same rate. Placing samples onto the ice bath would stop the reaction in either direction, allowing the user to load their unfolded protein samples onto the gel comb. If the protein was not unfolded via heating before being loaded onto the gel it would not show up as a

band later when imaging the blot, creating downstream consequences for students to experience.

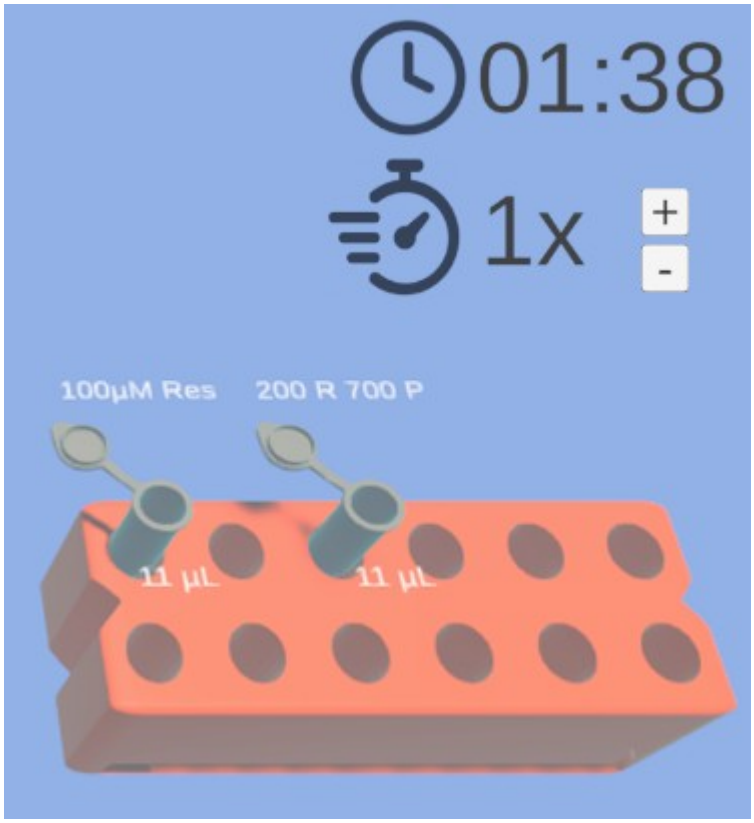


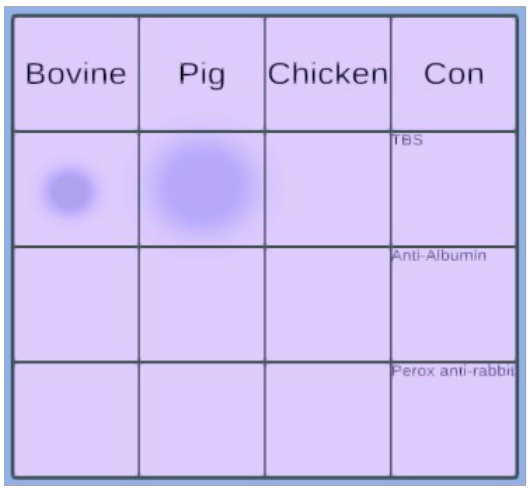
Figure Fehler: Verweis nicht gefunden-16 | Heating interface. Timer and time speed control that allow the user to monitor the current time and alter its speed, top right. Heating block representation with 11µL of each 100µM Resveratrol treatment and combined 200µM Resveratrol + 200µM Prednisolone treatments warming.

2.4.3 BMSC339

The immunoprecipitation stage of the BMSC339 immunochemistry laboratory was similar to the western blots in the previous laboratories, and such minimal new changes were needed to adapt it to the new context. The immunoblotting stage however, presented results in a new manner. In the immunoblotting stage, proteins and antibodies were pipetted directly onto a nitrocellulose membrane, and as such the resulting colour formation was in the form of circular “dots” and not bands as in previous western blots. This required a new method of pipetting solutions onto a nitrocellulose membrane while recording the locations and volumes of each instance, so that the dot output could appear correctly.

2.4.3.1 Pipetting onto the nitrocellulose membrane

A



B

```
GameObject NewSpot = Instantiate(Spot, Location, Spot.transform.rotation);

NewSpot.transform.localScale = new Vector3(New_Spot_Fluid_Data.Volume / 5f, 0, New_Spot_Fluid_Data.Volume / 5f);
```

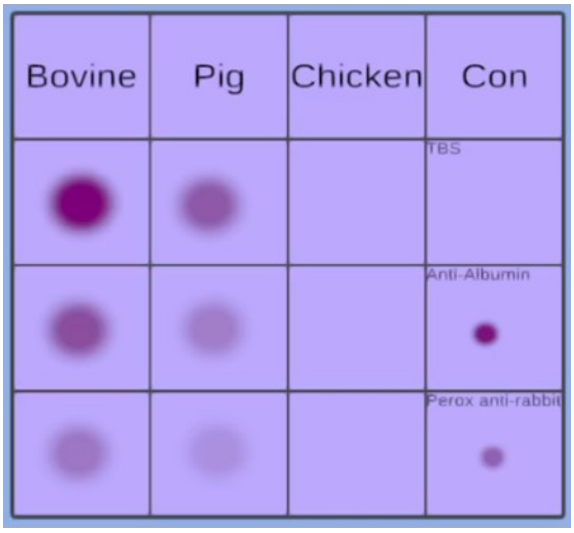
Figure Fehler: Verweis nicht gefunden-17 | Depositing fluids onto nitrocellulose membrane. a, Virtual nitrocellulose membrane in purple, with 5 µ and 10 µL dots below Bovine and Pig respectively. **b,** Script responsible for instantiation and scaling of new dots. Note the .localScale variable, that is dependent on the .Volume of the fluid.

In all previous virtual laboratories, all pipetting had been done from one container to another, in the case of test tubes, Eppendorf tubes, or the wells in an SDS-PAGE gel. The first new mechanic created was that of allowing students to pipette onto the nitrocellulose membrane. This was achieved by recording the location of the pipette tip when a volume was pipetted down and creating an opaque circle on the membrane as a visual representation of a wet dot. The radius of the dot was also based upon the volume of the fluid deposited (Figure 2-17 B).

2.4.3.2 Gradual colour formation

After secondary peroxidase conjugated antibodies were bound to other proteins on the membrane, the final step was to apply a 4-chloronapthol substrate solution that the peroxidase would catalyse into a coloured solution, visible to the naked eye. In the physical laboratory, this could be watched live by students over the course of roughly an hour, over which the dots of the dot blot would gradually become visible. In developing this laboratory, it was sought to replicate this experience instead of having the dots instantly appear. This realism would be important to students understanding of the gradual catalysis of the reaction over time. To recreate this, each dot that had secondary antibodies successfully bound was recorded in a list. When the membrane was incubated with the substrate solution, each of these dots slowly regained opacity over time (Figure 2-18 A). The colour of the dots was also changed to purple as they reappeared, representative to the chromogenic product generated in the wet lab.

A



B

```
newColor.a = MaxOpacity / Appear_Time * Time_Passed;

r.material.color = newColor;
```

Figure Fehler: Verweis nicht gefunden-18 | | Virtual dot blot results. a, Screenshot of final result from within the virtual laboratory tool, obtained by following laboratory protocol. Dots of the albumin species the left correspond to 10 µL of undiluted, 1/10 dilutions and 1/100 dilutions respectively. Note the positive response for both anti-albumin primary and anti-rabbit secondary antibodies in the control column. **b,** Script responsible for colour reappearance. This code would be executed every frame as the chloronapthol was catalysed to coloured product, gradually changing the colour of the dot.

2.5 Live implementation for learners

Each of the three virtual laboratories was provided to students to access during their timetabled practical classes, though students that did not complete their work in the allotted time were able to access the virtual laboratory outside of teaching hours. Students were able to use the virtual laboratory by navigating to a website where it was hosted online. This website was accessible to students on any device capable of running a modern internet browser, though was not available on mobile devices.

At the beginning of each online laboratory session, students were given a 15 to 30 minute tutorial on how to use the virtual laboratory. This included a live screenshared demonstration of the virtual laboratory, with a walkthrough on what each interface element was and how they could be used to interact with the virtual benchtop. Students were shown how to set the volume of a pipette tip, how to pipette fluids between containers, discard pipette tips, and any other scene specific tools in their respective laboratory. Care was taken to avoid demonstrating to students the correct protocol methodology, to prompt them to decide how to execute it correctly on their own. Students were also prompted to ask questions about any elements they did not understand, so that they could use the virtual laboratory immediately themselves when the tutorial ended.

Each of the cohorts using the virtual laboratory did so for the entire duration of two three hour long practical laboratory sessions, in the form on online Zoom calls. Students were able to ask questions of the teaching staff relating to the material during this period and were guided to complete the relevant practical within the virtual laboratory. Throughout their use of the virtual laboratory, they were encouraged to screenshot and save their outputs, successful or not. They could then place these within their electronic notebook and comment on the data. Each of the courses also had additional questions for the students to answer relating to the practical, which were used in the grading of their notebook.

2.6 Data collection

Data collection was conducted via questionnaires for both students and instructors following their interactions with the virtual laboratories. These questions aimed to

elucidate the nature of the students' interactions with the virtual laboratories, to analyse the success of the laboratories as learning tools.

Initially it was also planned to collect data measuring how students interacted with the virtual laboratory from within the virtual laboratory itself, such as the duration of their use of the laboratory, and the actions that they took within it. This method of data collection was not implemented into the virtual laboratory however, due to time and technological constraints.

Another potential source of data was the practical examination results from assessments pertaining to skills acquired in the laboratory. These results could have been collected from both cohorts that had access to the virtual laboratory and those that did not, and comparisons drawn between them. This also proved unviable, as ethical barries prevent the collection of student data, and confounding factors between cohorts would have rendered the data invalid. This was instead done by the inclusion of a question pertaining to these examination results in the questionnaires send to teaching staff.

2.6.1 Teaching staff questionnaire

Identical questionnaires were sent to each teaching staff overseeing each of the three laboratories after they had observed their students interacting with the virtual laboratories during online practical time.

The questions used in this questionnaire aimed to provide insight into the difficulties of providing an online practical experience without a virtual laboratory, and if the virtual laboratory they had used had helped to address these challenges. Additional questions were also asked to provide room for instructors to communicate their satisfaction with the virtual laboratory and provide any additional miscellaneous comments they may have had. The final list of questions below was used to collect this data, with a recommended response length of one to two sentences included.

1. Did you previously implement the course online, and if so, what were the challenges you faced?
2. Did your students benefit from using the lab?

3. What skills did the virtual lab succeed in teaching your students, and what skills was it unable to teach?
4. From assessment results, did you find any differences in knowledge between cohorts that had used the lab and cohorts that did not?
5. Would you consider using the virtual laboratory again in the future?
6. Do you have any addition comments that were not covered by the previous questions?

2.6.2 Student questionnaires

After each of the virtual laboratories were used by students in a practical class, data was collected from the students in the form of questionnaires. A voluntary survey was sent to each of the three cohorts via their online learning portal, aiming to investigate their outcomes from the virtual laboratory and opinions on its usefulness. Students in BCMB3X01 and BMSC339 were also required to answer questions relating to the virtual laboratory as a part of their assessed laboratory work, with details of each of these tasks below.

2.6.2.1 BCMB3X01 Gene and Genome Regulation Student Questionnaire

In the first course students were asked to fill out a voluntary survey after using the virtual laboratory. This survey asked students how long they had used each of the full and abridged versions of the virtual laboratory and if they had obtained successful outputs with either. Students were also asked to answer six additional statements, that were responded to with the use of a Likert scale.

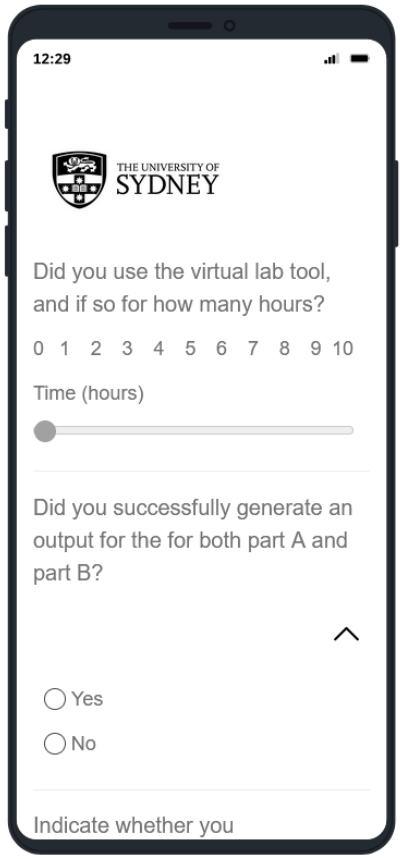


Figure Fehler: Verweis nicht gefunden-19 | Student survey graphic. Students were sent a hyperlink to the survey via their online learning portal that would allow them to take the survey.

The six statements chosen were as follows:

- I enjoyed using the simulator
- I would be able to complete the activity without any instructor help
- I was able to troubleshoot my own mistakes in the simulator
- It helped me to remember the steps involved in western blot
- It helped me to understand the purpose of each step of a western blot
- I would be more comfortable troubleshooting a real western blot

This survey aimed to collect data on student enjoyment of the laboratory experience and student confidence in performing the technique, both important metrics in assessing the success of a laboratory. Additionally, questions pertaining towards how the students interacted with the virtual laboratory and the methods they used to obtain their results are asked. This was done to investigate whether the virtual laboratory was causing students to adopt a learning approach that followed the active learning outcomes of interactive engagement and inquiry-based learning.

2.6.2.2 BCMB3X03 Biochemistry of Human Disease Student Questionnaire

In this course, students were asked to answer a modified survey after using virtual laboratory. The first two questions remained the same as the previous survey asking how long and how successful the student's use of the virtual laboratory was, though the Likert scale response questions were changed. The questions regarding instructor help and the memorisation of the steps of a western blot were removed, in favour of another question aiming to investigate how students interacted with the virtual laboratory. These steps were removed to shorten the length of the survey to encourage higher participation, and the final list of questions for this cohort can be seen below.

- I enjoyed using the virtual laboratory tool.
- I improved my blot output by successive uses of the tool.
- When using the tool, I just quickly tried different options until I had achieved a desired result.
- When using the tool, I had to consider why I was not getting the desired output in relation to the biochemical mechanisms at work and alter my workflow appropriately.
- I would now be more confident performing and evaluating the results of a real western blot after using the simulator

An additional optional question was appended to the survey, allowing students to provide any additional feedback on the virtual laboratory.

Students in this cohort were also assigned a compulsory question regarding the virtual laboratory as a part of their laboratory report. Each student was randomly allocated a one of three questions, asking how one of either the sample preparation, gel transfer, or blotting stages could be made more realistic or user friendly. This prompted students to consider any aspects of the virtual laboratory that differed from the physical laboratory, and how it could be modified to be more realistic. The question not only tested student's theoretical knowledge on the principles of the protocol, but also provided valuable feedback on how the virtual laboratory could be improved in future.

2.6.2.3 BMSC339 Cellular Regulation Student Questionnaire

The voluntary survey sent to students in this cohort was identical to the one sent to students of BCMB3X03, excepting the first question asking how many hours the students had used the virtual laboratory for. This question was changed from a scale of 0-5 hours to a 0-10 hour scale, owing to students spending more time with the virtual laboratory in this course.

Students of this cohort were also the following question as a part of their laboratory report, providing long form responses on various aspects of the virtual laboratory that could be improved, in addition to information on overall student satisfaction.

'Providing some helpful feedback on the operation of the simulator in your submission. This can be suggestions for how the 'game play' could be improved, how well/poorly you think the simulator serves as a data-generator, and what (if anything!) you have learnt from using the simulator.'

3 Results

3.1 The Virtual Laboratories

3.1.1 BCMB3X01 Gene and Genome Regulation Virtual Laboratory Program

For the BCMB3X01 Gene and Genome regulation course, the aim was to develop an online virtual laboratory for students to conduct a western blot experiment. This would require students to determine sample protein concentrations, dilute and load samples of appropriate volumes onto an SDS-PAGE gel, transfer the gel to a blot membrane and finally execute the necessary steps of a western blot to obtain an exposed blot image.

The core goal of the program was to allow students to experiment with the laboratory as they saw fit, and experience consequence as they would in the physical laboratory, in the form of the results obtained. The virtual laboratory tool developed accomplished this, with customised data generated for students based upon their actions in the program.

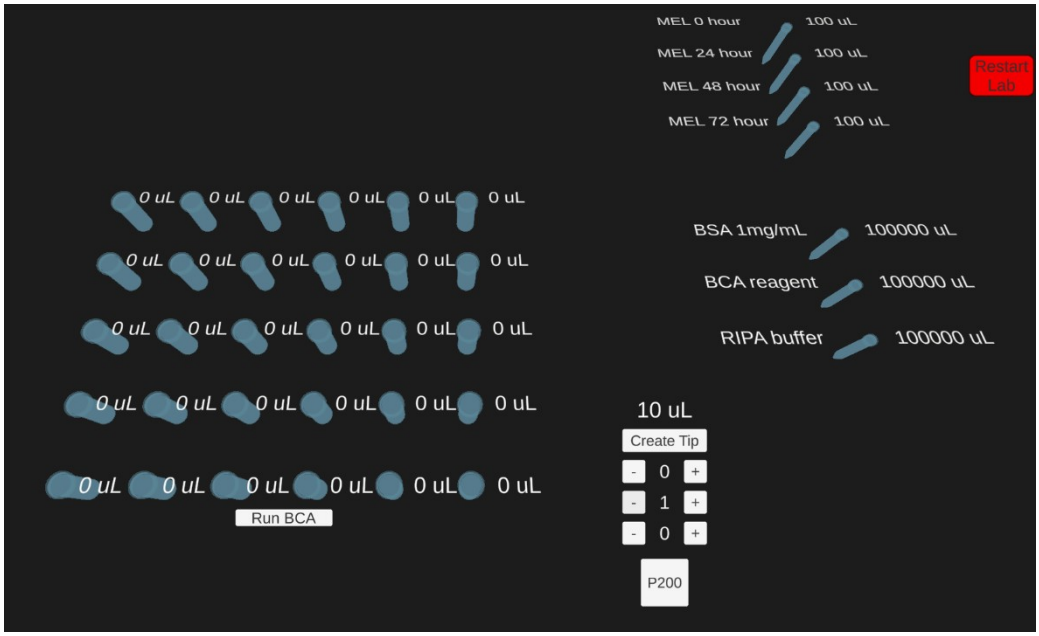


Figure Fehler: Verweis nicht gefunden-1 | BCMB3X01 Protein quantification scene. In this scene, students were able to create a pipette tip of a chosen volume using the controls bottom right, and use this to dilute any of the reagents in the upper right as necessary for the BCA assay.

The program began with a protein quantification step as seen in Figure 3-1, where students would be required to determine the concentration of four MEL cell treatments of distinct durations. Students would have access to each of the samples, 1 mg/mL BSA, BCA reagent, and RIPA buffer. Using these reagents, they would be able to perform a serial dilution of all protein samples in the supplied array of tubes and obtain a tabulated output result of the plate reader (Figure 3-2).

Row/Column	A	B	C	D	E	F
1	1.9	1.9	1.164	0.665	0.39	0.264
2	1.59	0.908	0.525	0.328	0.225	0.176
3	1.356	0.779	0.461	0.288	0.201	0.158
4	1.9	1.9	1.9	1.47	0.82	0.471
5	0.571	0.348	0.232	0.177	0.155	0.121

Figure Fehler: Verweis nicht gefunden-2 | BCA Assay output. An absorbance reading obtained by performing a serial dilution of 1 mg/mL BSA in row 1, and a serial dilution of each of the MEL samples in row 2-5.

Once the absorbances were obtained, students would be required to calculate the protein concentrations of the MEL samples, that would be required for the next step. To do this, they would need to construct a standard curve from the protein standards of known concentration and use this to estimate the protein concentration of their MEL samples. These data analysis steps would be performed in Microsoft Excel or another spreadsheet and graphing program. By pressing the 'Go to loading' button after the concentrations were calculated, students were able to proceed to the next stage.

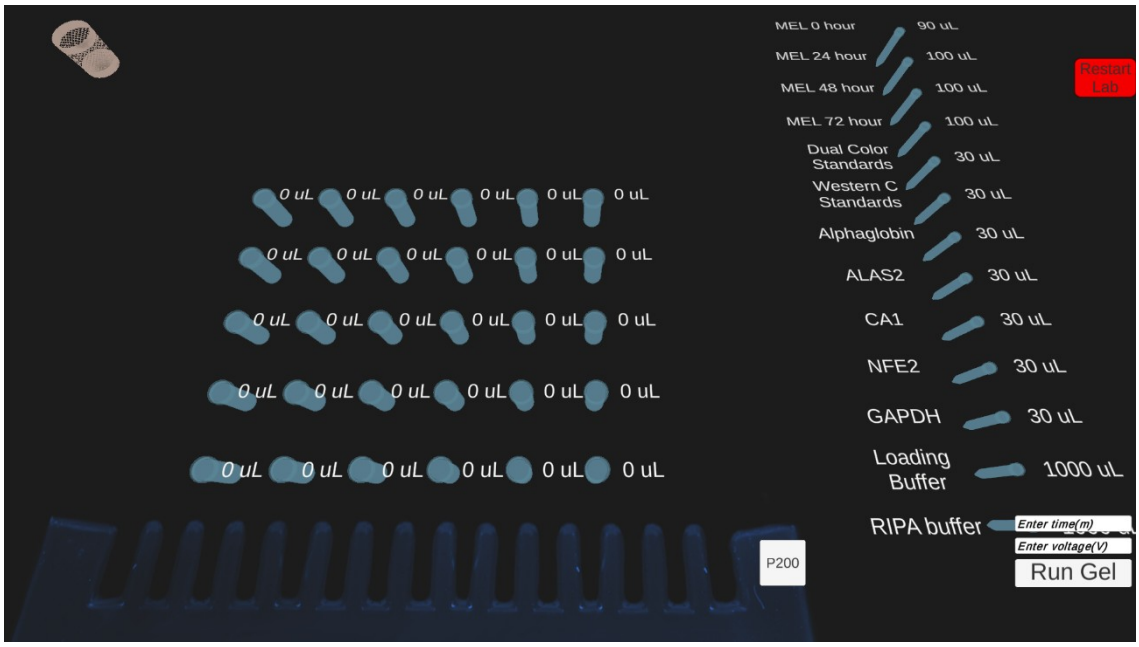


Figure Fehler: Verweis nicht gefunden-3 | BCMB3X01 loading stage. In this stage, students would be able to load each of the wells of the 15 well gel using the pipette and mixing array. The data contained in each well was used to generate the output Figure 3-4.

At the loading stage, students were able to load the wells of the 15-lane SDS-PAGE gel with their samples and additional protein standards as they saw fit (Figure 3-3). Students were supplied with the same protein samples, Dual colour and western C standard, controls for target proteins, and loading and RIPA buffers. The pipetting tool and an array of tubes were also available to students to allow dilution and mixing of samples. The laboratory instructions specified that 15 µg and 45 µg of each MEL sample should be loaded into replicate wells, so it was important students completed their concentration calculation correctly to be able to achieve this. The supplied protein controls were of known concentration and were able to be loaded directly onto the gel once mixed with loading buffer. Each well held a maximum of 15 µL and did not allow students to overfill.

Once the student was satisfied with the mixtures that they had loaded into the wells, input fields allowed them to specify the time and voltage for which the gel was run. A time of 50 minutes and a voltage of 150v would yield optimal results, and deviations from this would cause the gel to run a corresponding distance, for example if the gel

was run for only 20 minutes, all proteins would not yet have reached the halfway point on the gel.

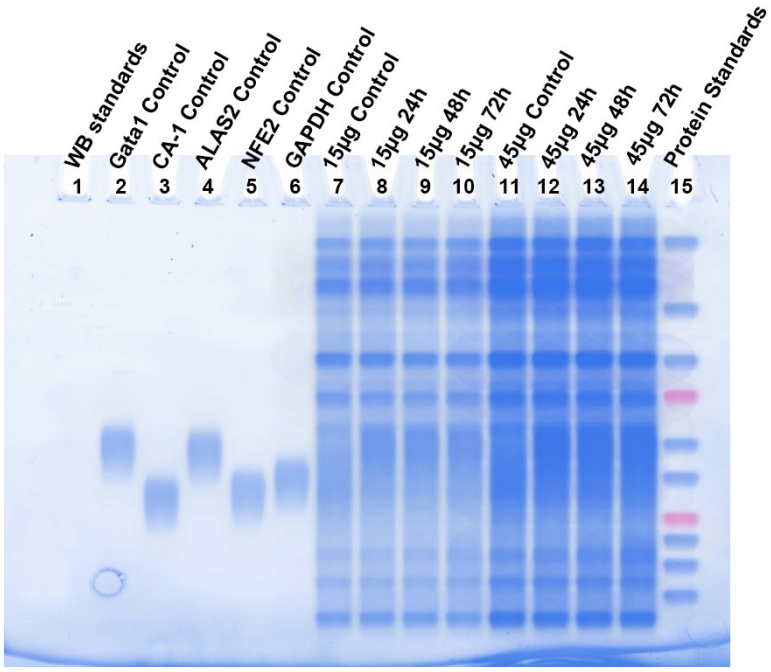


Figure Fehler: Verweis nicht gefunden-4 | Gel loaded according to laboratory protocol. This gel was obtained by loading the wells in accordance with the laboratory protocol, and would allow students to probe for any control protein.

If students were satisfied with the outcome of their SDS-PAGE gel (Figure 3-4), they could proceed with an unstained copy to the transfer step. If the results indicated a problem with the results (for example, too much or too little protein sample had been loaded), they could return to the loading stage to reattempt it, which could be repeated as many times as required.

In the transfer stage, students were required to choose the order of their cassette sandwich, via the selection of parts from a six dropdown fields (Figure 3-5). Once an order was chosen, the 'Transfer' button would take students to the blotting stage. If an incorrect order of the western blot cassette components had been chosen, students also had the option to return to this step.

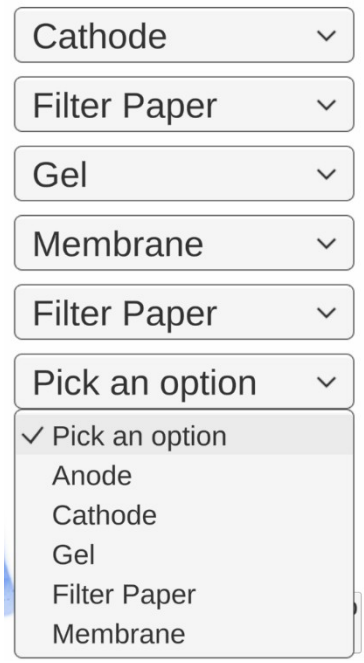


Figure Fehler: Verweis nicht gefunden-5 | Transfer cassette order. The transfer cassette order was chosen by a series of dropdown fields. The correct order would be accepted in either direction.

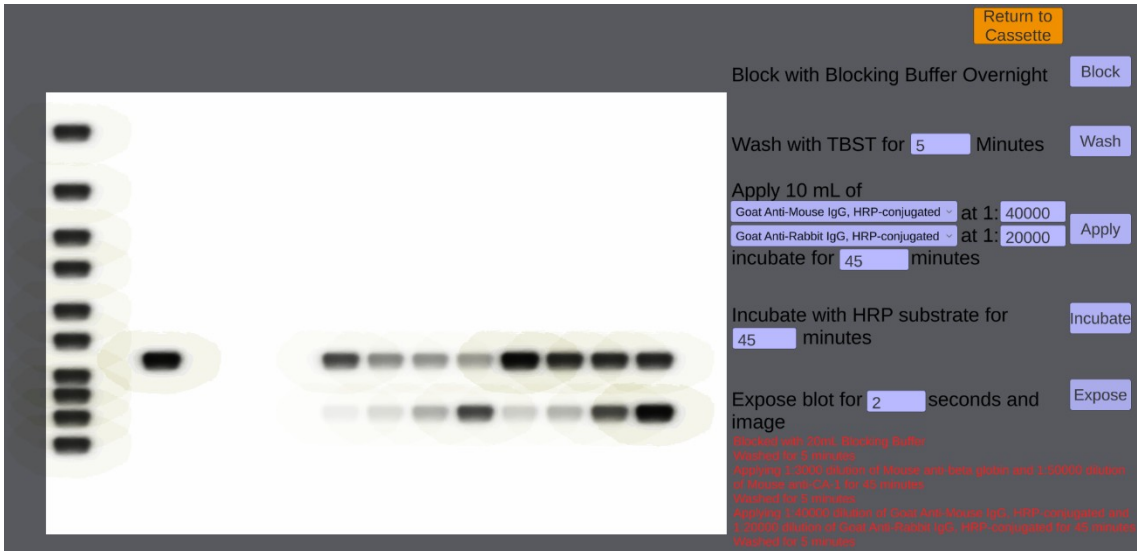


Figure Fehler: Verweis nicht gefunden-6 | Successfully imaged blot. Blot probed for CA1 and beta globin.

In the blotting stage, students would be able to blot, wash, apply antibodies to, or incubate the membrane with HRP substrate. Additionally, students could select the ratio at which their antibodies were applied, and the duration of the incubation steps. All these choices affected the intensity of the final imaged output, which could be obtained by pressing the 'Expose' button with a duration specified. A correct order

would be to block the membrane, wash the membrane, apply primary antibodies, wash the membrane, apply secondary antibodies, wash the membrane, and then finally incubate the membrane with HRP substrate. If done at the correct concentrations and durations, the blot could be successfully imaged, as in Figure 3-6. Deviations from the protocol in either the concentration or duration fields could result in little or no result, if for example the ratio of an antibody was too low, or the HRP substrate was not incubated for long enough. These changes would scale the opacity of the imaged blot accordingly and halving any of these parameters would result in a lighter blot image.

If the order of the steps was chosen incorrectly, such as the secondary antibody being applied before the first, then no result would be obtained. This is in accordance with the output that would be obtained in a real laboratory executed the same way. As such, it allowed students to experience consequence and iteratively examine their results, reflect on their decisions, modify their approach and improve their outputs within the virtual laboratory.

3.1.2 BCMB3X03 Biochemistry of Human Disease Virtual Laboratory Program

The 'BCMB3X03 – Biochemistry of Human Disease' course virtual laboratory aimed to provide students with an online experience of a Western blot experiment. The goal of this virtual laboratory was also to allow students to experiment and experience consequence, while they investigated the effects of drug treatments on cell survival.

The first stage involved both sample preparation and gel loading, the objective being to load the wells of the SDS-PAGE gel with the unfolded protein samples in each treatment. Students were supplied with eight distinct cell treatments, two levels of each a resveratrol, prednisolone, and combined treatments, and both a media and DMSO control (Figure 3-7). The protocol instructed students to add 4 µL of loading buffer to each sample, and then place the samples on the heating block to unfold the proteins within, as only unfolded proteins would appear on the gel in the next stage. One third of the protein in each sample would become unfolded per minute, and this process could be sped up by the use of a timer. Once sufficient time had passed, students could place their samples in an ice bath to halt this process and begin to load

the wells of the gel. Samples placed on the bench instead would begin to undergo a reversal of this unfolding and return to their folded state.

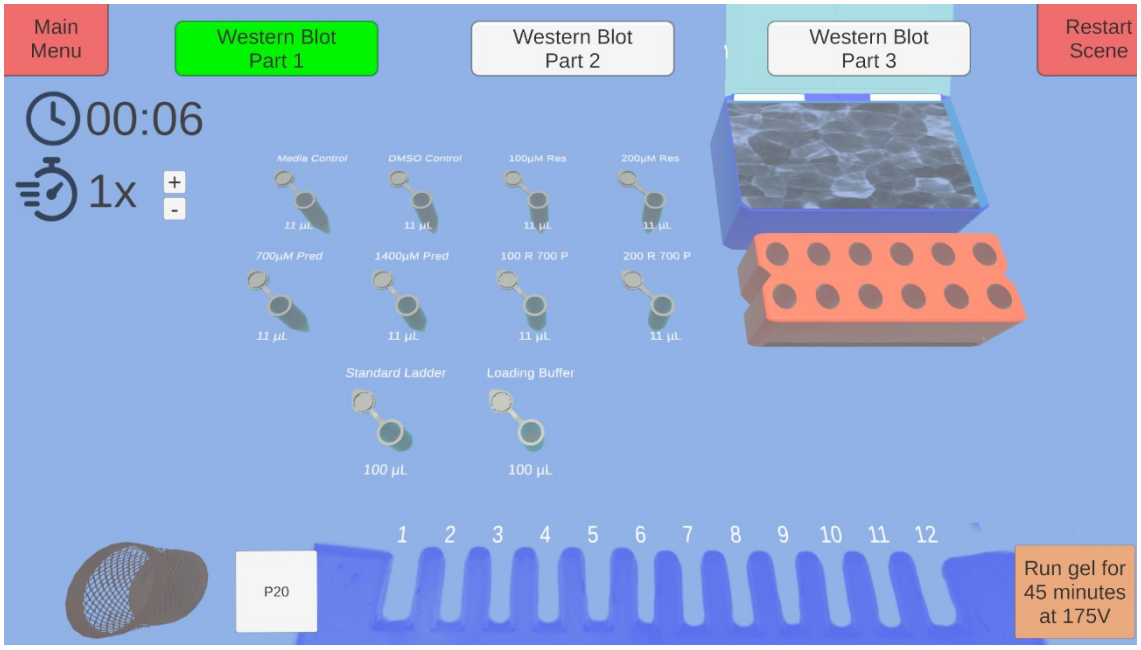


Figure Fehler: Verweis nicht gefunden-7 | BCMB3X03 loading stage. The first stage of the BCMB3X01 Western blot. In this stage students could drag the Eppendorf tubes onto both the orange heating block or ice bath above.

Once the wells were loaded, the 'Run gel for 45 minutes at 175V' button could be pressed to proceed to the next stage.

At the next stage, the dye-front would be visible at the bottom of each lane that had been loaded with a protein sample, and the order of the transfer cassette could be chosen as in BCMB3X01 (Figure 3-8). Once chosen, the transfer button would take users to the final blot stage.

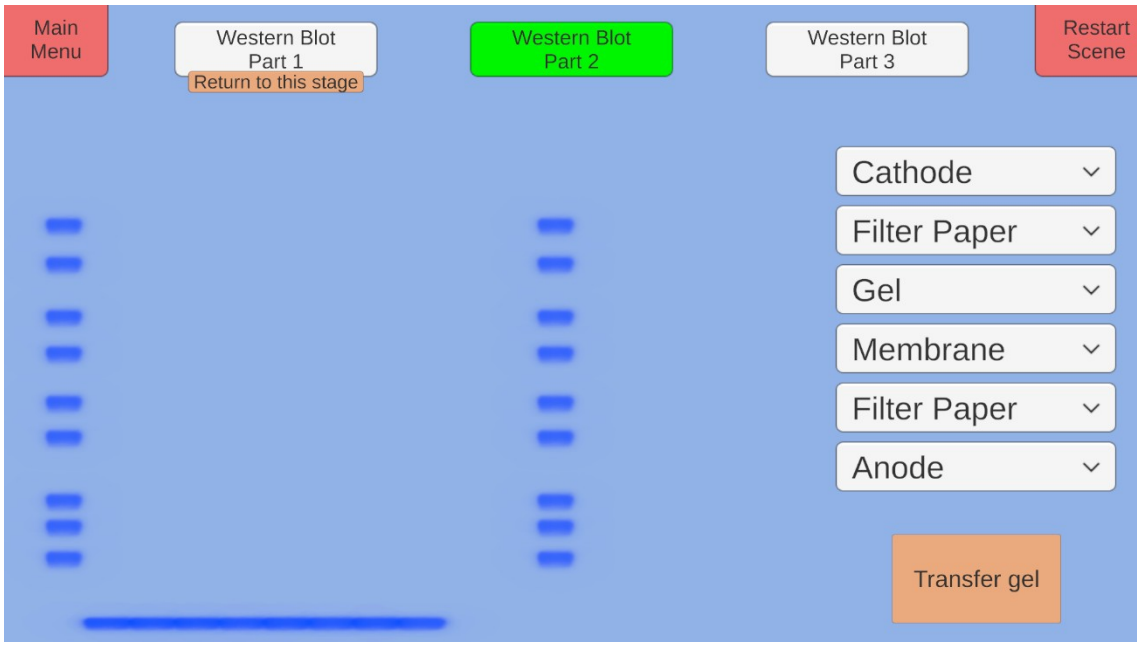


Figure Fehler: Verweis nicht gefunden-8 | BCMB3X01 SDS-GEL image. SDS-Page gel output, with dyefront visible at the bottom of the gel and standard ladders on each side. To take this gel to the next stage, students would select the cassette order on the right and then press the 'transfer gel' button. To repeat this process, students would press 'return to this stage' below the part 1 header.

The blot stage was a varied version of the one present in BCMB3X01, omitting the antibody ratio and incubation duration inputs, but adding additional antibody selections and blocking and washing solutions. If the process was executed correctly students could obtain their own blot image, otherwise the scene could be restarted and attempted anew. This final output could then be screenshot and placed in the student's laboratory notebook (Figure 3-9).

Figure Fehler: Verweis nicht gefunden-9 | BCMB3X01 blot output. Successfully imaged gel probed for Cas-3 and alpha tubulin. The upper bands correspond to the alpha tubulin(50 kDa) loading control, with Cas-3(32 kDa) beneath. In rows 5 – 9, cleaved Cas-3(10 kDa) can be visible, a marker for apoptosis.

3.1.3 BMSC339 Cellular Regulation Virtual Laboratory Program

The 'BMSC – Cellular Regulation' virtual consisted of two sections, an immunoblotting stage, where a dot blot protocol would be used to investigate antibody affinity, and an immunoprecipitation stage, where a bovine serum sample would be tested for the presence of albumin.

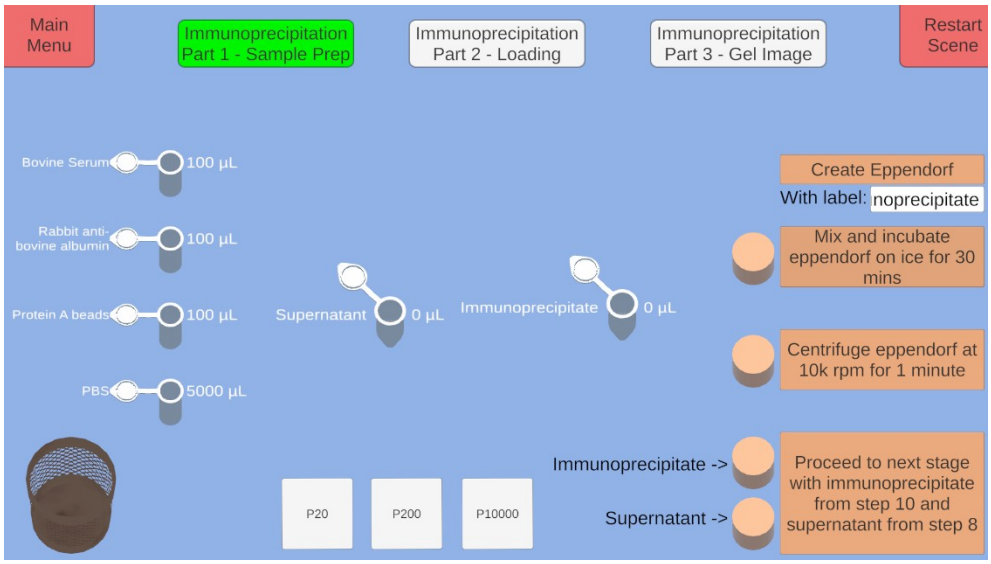


Figure Fehler: Verweis nicht gefunden-10 | BMSC339 Sample preparation stage. In this stage, students divided supplied bovine serum into supernatant and immunoprecipitate tubes, which were then placed on the corresponding platforms in the bottom right to be taken to the next stage. If samples were not prepared correctly this would have downstream consequences in the gel results.

The immunoprecipitation stage began with a sample preparation scene, where students would separate a bovine serum sample into a supernatant sample and an immunoprecipitated pellet sample to take to the next stage (Figure 3-10). First, a serum sample would be diluted with PBS and then combined with protein A conjugated polymer beads. This solution would be incubated on ice and centrifuged, with the supernatant retained and the pellet discarded. This would act as a pre cleaning step to prevent nonspecific binding of the polymer beads to the bovine serum. Next, the remaining supernatant would be mixed with anti-albumin antibodies and a bacterial protein conjugated to polymer beads that was a binding partner of the antibody, and this mixture centrifuged. The supernatant of this solution could then be removed to a new Eppendorf, and the pelleted material resuspended. These two Eppendorf tubes could them be placed on designated locations that would allow them to be brought with the user to the loading stage.

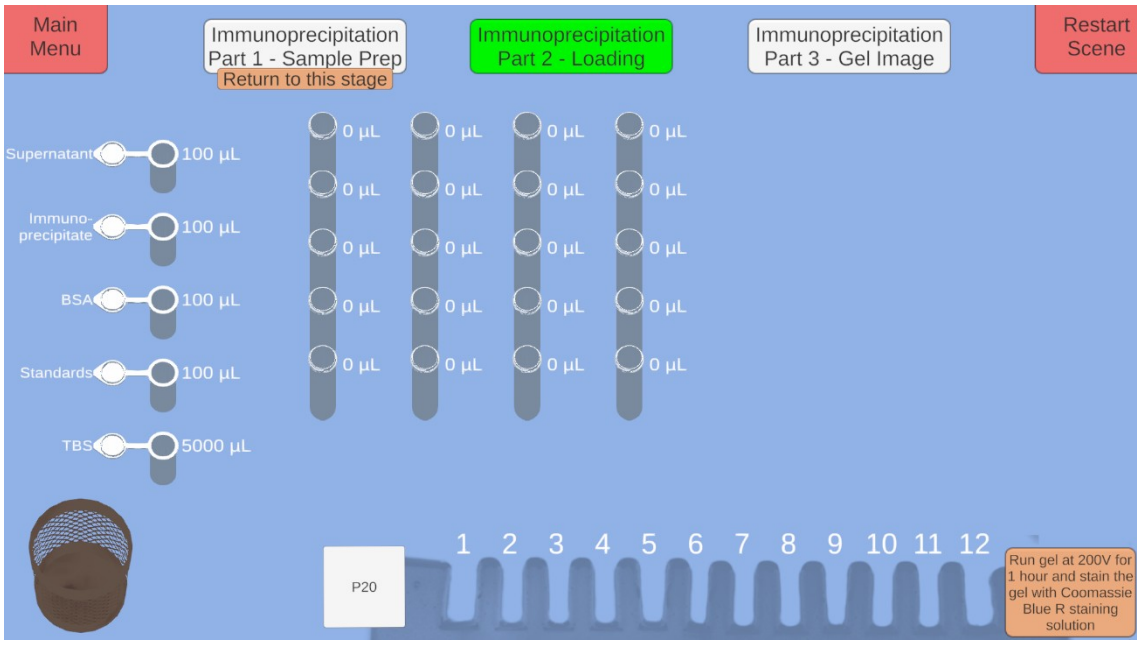


Figure Fehler: Verweis nicht gefunden-11 | BMSC Gel loading stage | In this stage, students were able to load the wells on an SDS-PAGE gel as desired using their previously prepared samples and BSA.

The loading stage was similar to previous loading instances in the other laboratories, allowing students to mix and dilute their acquired samples and a BSA sample with TBS (Figure 3-11).

These samples and a standard ladder could then be loaded into SDS-PAGE gel wells, and the gel run at 200V for 1 hour and stained with Coomassie blue. A resulting stained gel image would then be acquired.

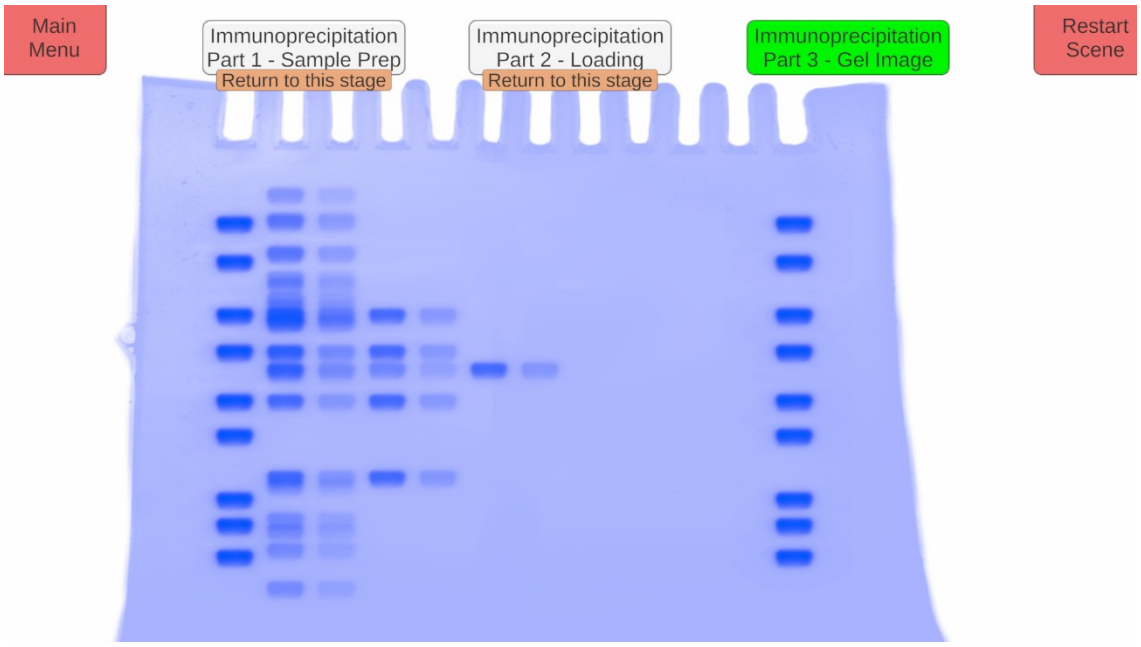


Figure Fehler: Verweis nicht gefunden-12 | BMSC SDS-PAGE results. A completed gel loaded as per the practical manual, with standard ladders in lanes 1 and 12, and 20 µL and 5 µL samples of supernatant(lanes 2-3), immunoprecipitate(lanes 4-5), and Bovine serum albumin(lanes 6-7). Note the presence of albumin(65kDa) in each lane.

From this gel image (Figure 3-12), students could then interpret if albumin was present in their bovine serum sample.

The immunoblotting scene allowed students to execute a dot blot protocol to investigate the affinity of anti-bovine-albumin antibodies for bovine, pig, and chicken albumin.

The protocol consisted of two scenes, divided at the point students would incubate the nitrocellulose overnight in primary antibody. In the first scene, students were presented with the three homologous antibody species, both primary rabbit anti-bovine albumin antibodies and secondary peroxidase conjugated anti-rabbit antibodies, water, TBS, and a 10% milk powder solution (Figure 3-13). These reagents could then be diluted and mixed as required using the pipette tips provided and an array of empty tubes, and then pipetted directly onto the nitrocellulose.

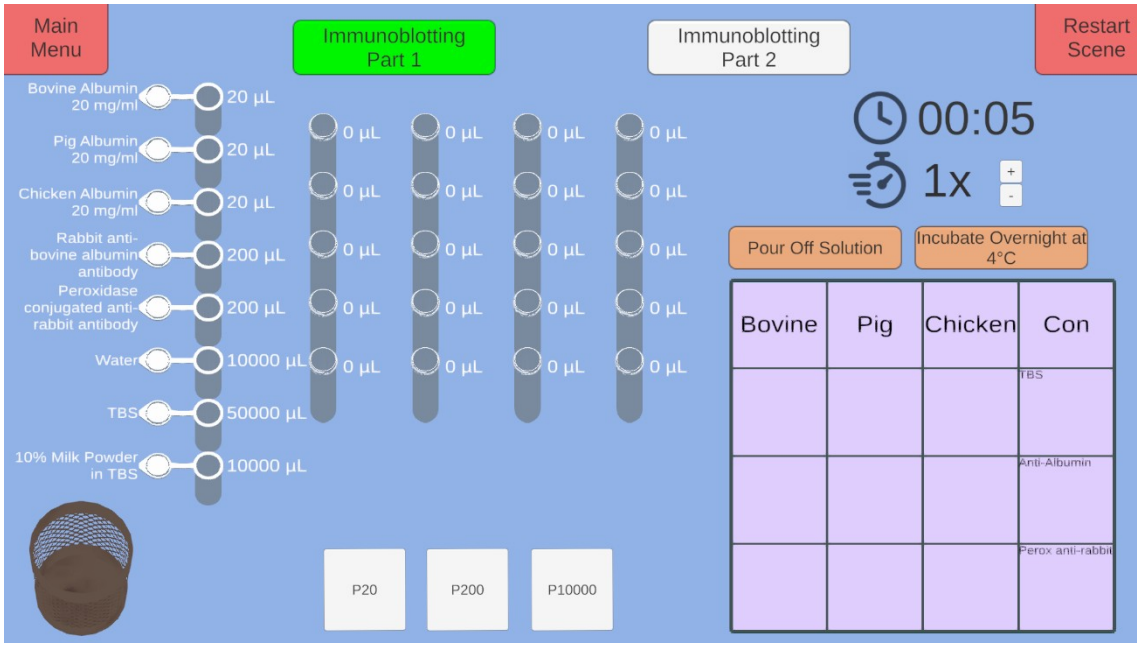


Figure Fehler: Verweis nicht gefunden-13 | BSMC339 Immunoblot Stage 1. Initial state of the immunoblotting laboratory, with reagents on the left, tubes for dilution centre, pipetting controls below and nitrocellulose bottom right.

The protocol specified that students should spot 5 µL of undiluted, 1:10, and 1:100 dilutions of each albumin species onto the nitrocellulose, in addition to 2 µL controls of TBS and both antibodies. Once all solutions were spotted onto the nitrocellulose, it would then need to be blocked in 10 mL of the 10% milk solution for 30 minutes to prevent nonspecific binding of antibodies directly to the nitrocellulose. After the milk solution was rinsed off the nitrocellulose, it could then be incubated overnight in the rabbit anti-bovine albumin antibody, taking students to the next stage.

In this stage, the aim was to bind the secondary antibody to the primary antibody affixed to the nitrocellulose, and then apply the chromogenic 4-chloronapthol stock that it would convert into a coloured product.

As such the reagents supplied to students in this stage consisted of secondary antibody, TBS, and 4-chloronapthol stock solution. Once the primary antibody was poured off the nitrocellulose, it could then successfully be incubated 10 µL of secondary antibody diluted in 10mL of TBS. After half an hour, the secondary antibody was bound, and the nitrocellulose could be rinsed. Application of the 4-chloronapthol substrate would begin the reaction, and coloured dots would gradually begin to appear on the membrane (Figure 3-14). The affinity of the anti-bovine albumin antibody for each of the reagents on the nitrocellulose could then be analysed according to the intensity of the colouring that appeared.

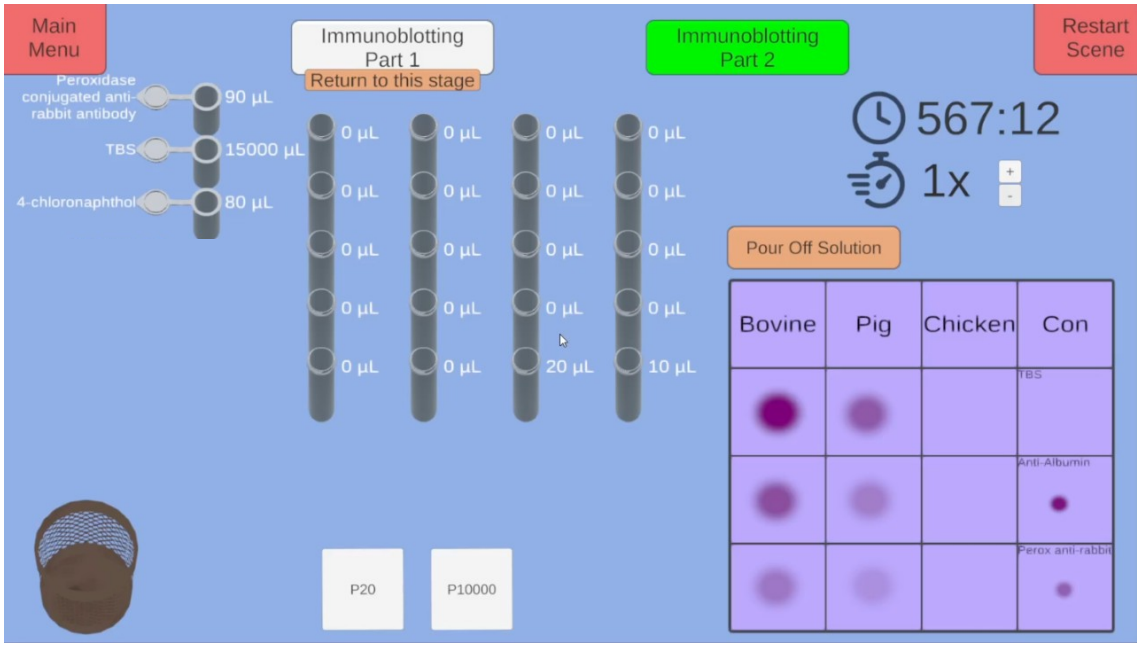


Figure Fehler: Verweis nicht gefunden-14 | Virtual dot blot results. Coloured dots representing formation of visible product appear on the nitrocellulose. The anti-bovine albumin antibody had a 100% affinity for bovine albumin, 50% affinity for pig albumin, and no affinity for chicken albumin.

3.2 Student feedback

3.2.1 BCMB3X03 – Gene and Genome Regulation

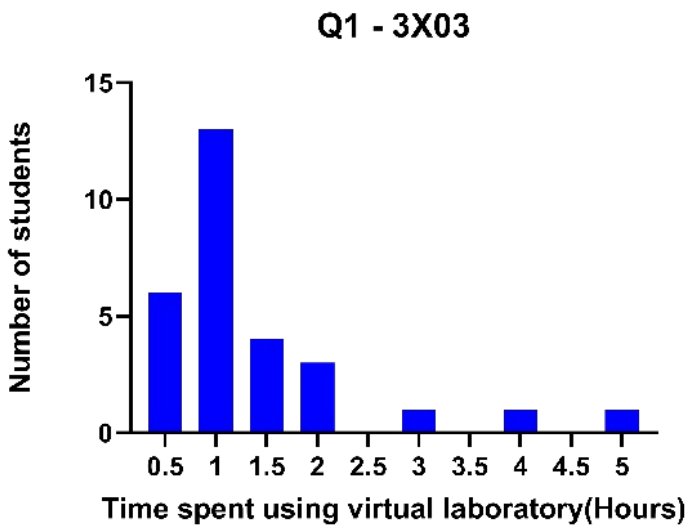


Figure Fehler: Verweis nicht gefunden-15 | Student Laboratory Use. Self-reported hours spent by students using the BCMB3X03 virtual laboratory. N = 19. Mean 1.38. SD 1.04. Results are positively skewed, with only 3 students using the tool for more than 2 hours.

Most students used the virtual laboratory for half an hour to two hours, as seen in Figure 3-5, though three students used it for 3, 4, and 5 hours respectively. All 30 of the students that answered the survey also successfully generated a blot output, indicating that they had used the virtual laboratory to completion.

In Figure 3-16, the student Likert responses to each of the chosen statements can be seen. Student enjoyment of the virtual laboratory and increased confidence in performing a physical version were both reported to be high, with average responses of 4.23 and 4.23 respectively (Figure 3-16 A/E).

Students also responded positively to statements that they improved their successive outputs of the virtual laboratory (Figure 3-16 B), though the next statement shows a clear bimodal distribution in responding to how this was achieved (Figure 3-16 C). Few students responded neutrally to the statement that they quickly tried different results, with roughly half agreeing and disagreeing each. In either case, students agreed that they had to consider the biochemical mechanisms at work between these iterations (Figure 3-16 D).

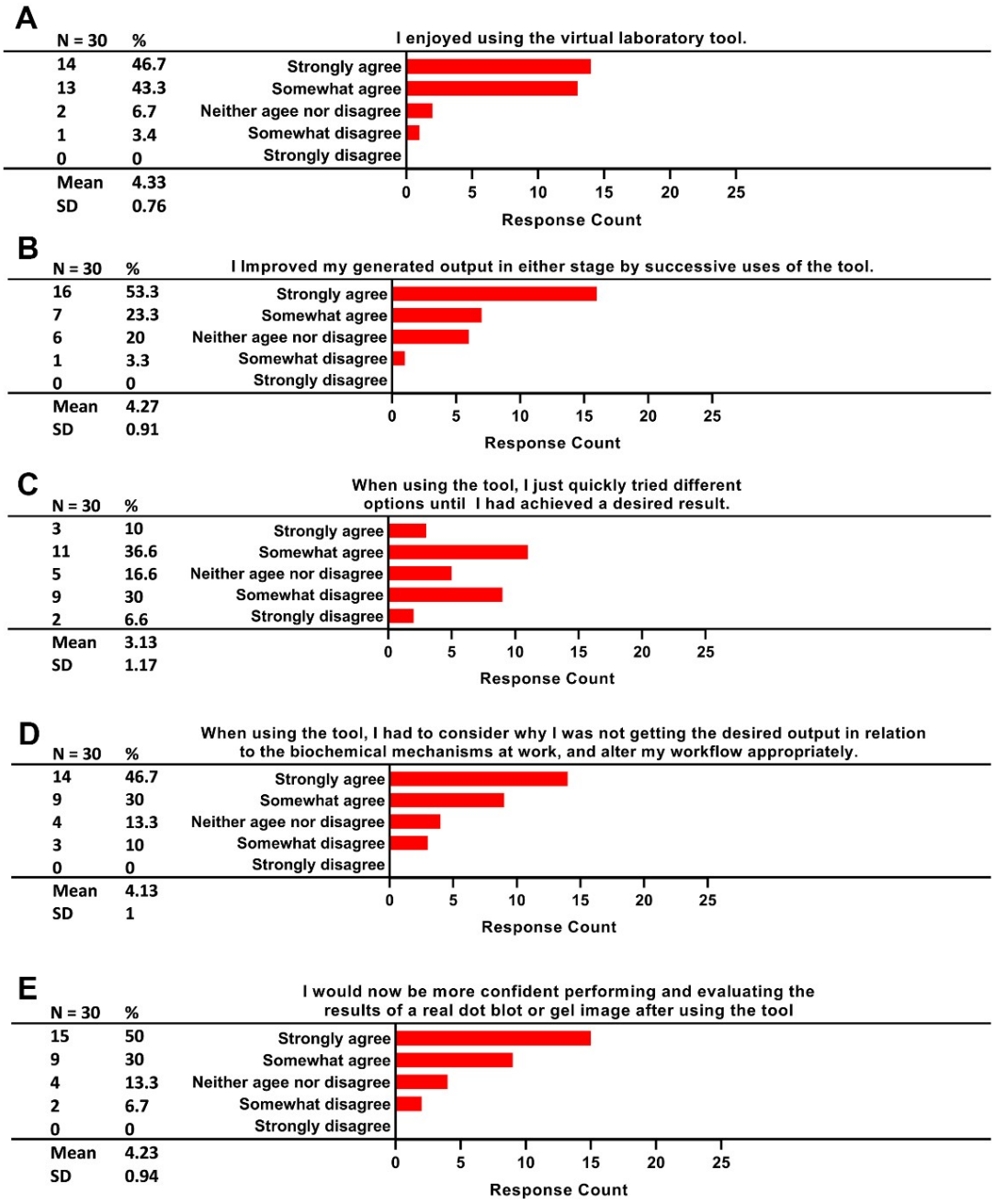


Figure Fehler: Verweis nicht gefunden-16 | Student Likert scale responses. Students were asked to rate their agreement with each statement on a Likert scale of 'Strongly agree" to "Strongly disagree". Responses were converted to numbers, with 1 representing a strong disagree response, and a 5 representing a strong agree. Students showed strong agreement (Mean > 4) with all statements except C, which instead displayed a bimodal distribution.

In the optional extended response question, students were asked to provide their own miscellaneous feedback, which was categorised into three buckets. All students responded positively, with a subset of students adding that a physical laboratory would have been better, or that the program could still use improvement (Figure 3-17). As a part of the student's laboratory report, they were asked to identify the main areas a section of the tool could be improved, assigned either the sample preparation, transfer, or blot stages (Figure 3-18). The most common areas that students identified as needing improvement were those of the graphics of the program or the addition of extra steps. Many students answering regarding the sample preparation stage also identified the mixing of solutions as a valuable addition.

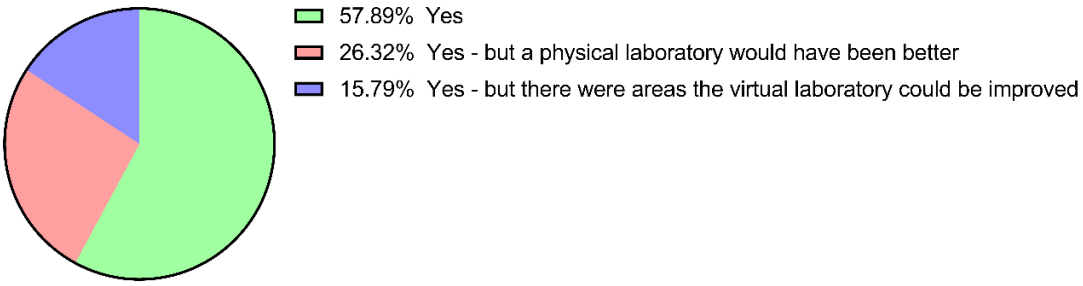


Figure Fehler: Verweis nicht gefunden-17 | Students view on the virtual laboratory as a learning resource. Students were given the option to input a response to the question "Do you think the tool was useful as a learning resource, and why or why not?" n = 19.

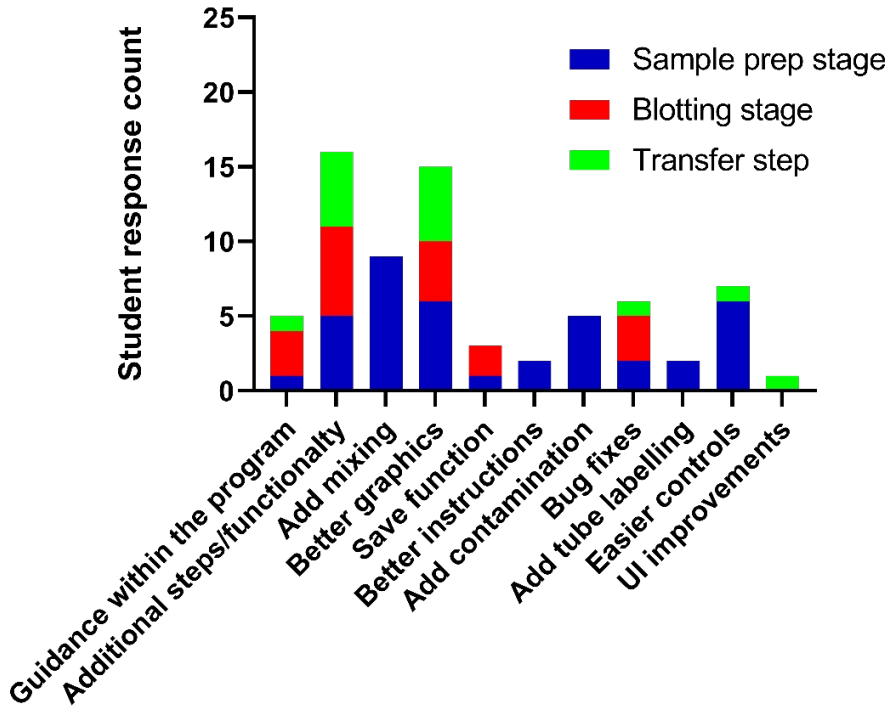


Figure Fehler: Verweis nicht gefunden-18 | Student Improvement Feedback. Students were asked to provide feedback relating to one of the three stages of the program. The responses were categorised based upon the main area of improvement specified. N = 71.

3.2.2 BMSC339

Students used the virtual laboratory tool in the BMSC339 course for an average of 4.39 hours, though there was a large sample variation, with six students using the virtual laboratory for over five hours (Figure 3-17). All but one of the 29 students that answered the survey successfully generated an output for both the immunoblotting and immunoprecipitation stages of the virtual laboratory. Students reported above average of enjoyment of the tool (Figure 3-18 C), with higher levels of confidence in their ability to execute a physical version of the laboratory (Figure 3-18 E). Students responded positively to statements both that they successively improved their output generated, and that they had to consider the biochemical reasoning behind their methodology (Figure 3-18 B/E). A bimodal distribution was seen in the statement that they quickly tried different options until a result was obtained (Figure 3-18 C).

In Figure 3-19, where students were asked to answer the question “Do you think the tool was useful as a learning resource, and why or why not?”, all of the students

expressed agreement. Half of these responses added additional comments, almost evenly split between that the physical laboratory would have been better, or that the virtual laboratory had areas for improvement.

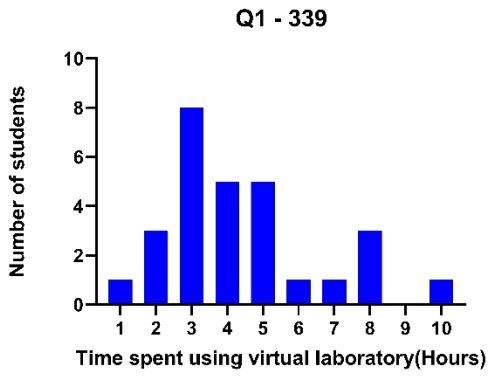


Figure Fehler: Verweis nicht gefunden-17 | Student Laboratory Use. Self-reported hours spent by students using the BCMB339 virtual laboratory. N = 28. Mean 4.39. SD 2.13. Results are positively skewed, with the majority of students using the virtual laboratory for 3-5 hours.

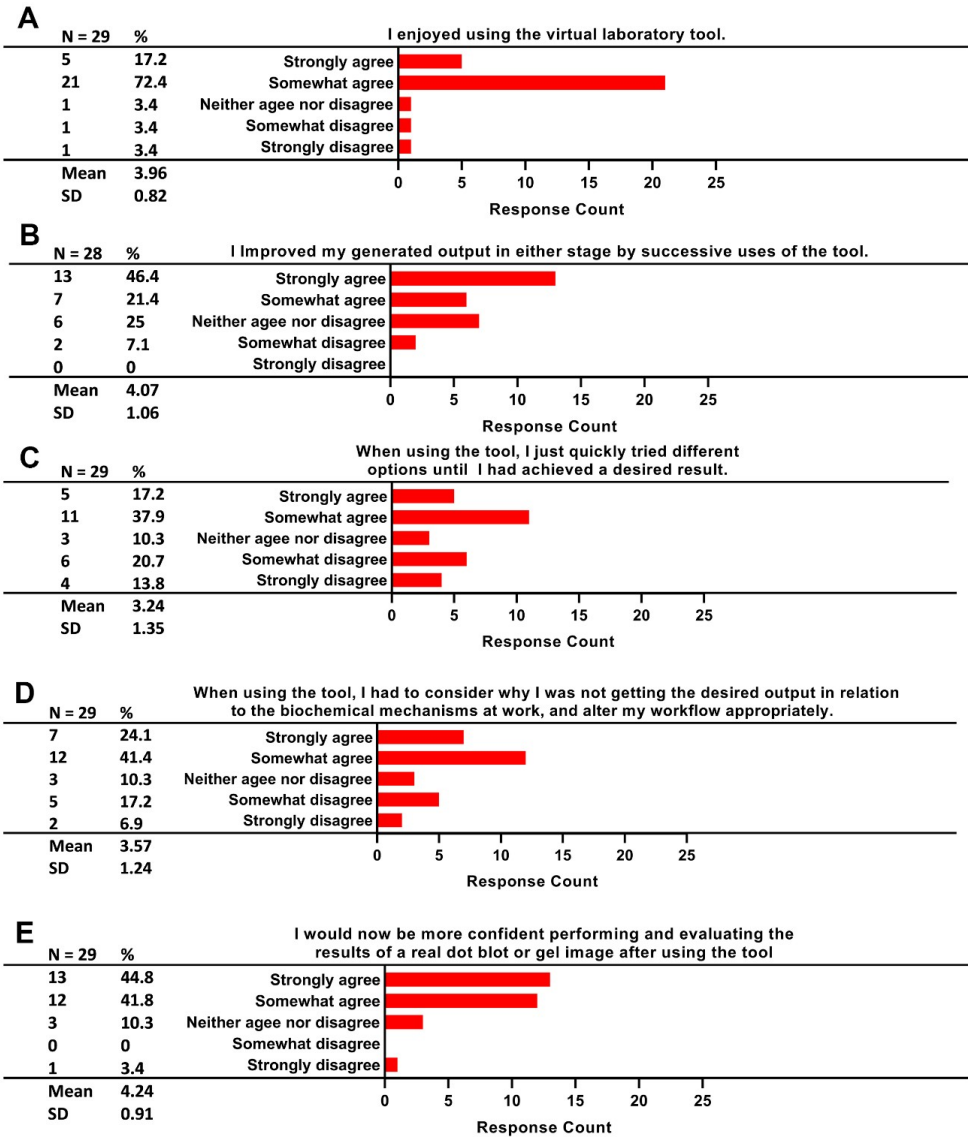


Figure Fehler: Verweis nicht gefunden-19 | BMSC339 Student Likert scale responses. Students were asked to rate their agreement with each statement on a Likert scale of 'Strongly agree" to "Strongly disagree". Responses were converted to numbers, with 1 representing a strong disagree response, and a 5 representing a strong agree. Students showed strong agreement with statements B and E (mean > 4), with a lesser agreement to statements A and D (3 < Mean < 4). Statement C yielded a bimodal distribution.

Students were additionally asked as a part of their laboratory report to provide specific feedback on where the tool could be improved, and responses were categorised in Figure 3-20. Responses recommending adding graphics to the centrifugation step or resulting pellet (29.11% of responses) or provide a clearer instruction sheet (27.84% of responses were the most commonly identified areas, with the addition of an undo

button (17.72% of responses) coming in third. The remaining responses were split among many smaller issues.

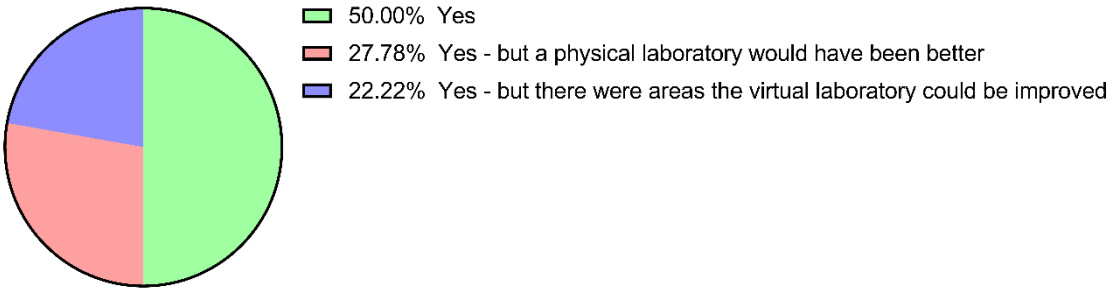


Figure Fehler: Verweis nicht gefunden-20 | Students view on the virtual laboratory as a learning resource. Students were given the option to input a response to the question "Do you think the tool was useful as a learning resource, and why or why not?" n = 19.

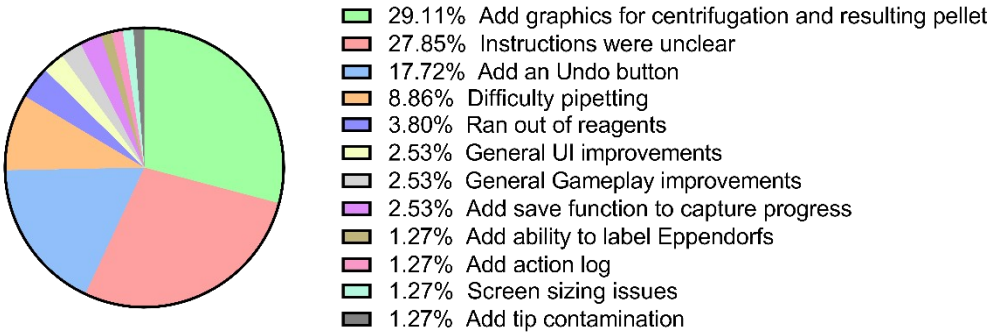


Figure Fehler: Verweis nicht gefunden-21 | BSMC Feedback survey. Students were prompted to provide any feedback on the virtual laboratory, which was then categorised based upon the main issue brought up in each response. N = 79.

3.3 Instructor feedback

A set of feedback questions was sent to each of the unit coordinators after participation in the virtual laboratory as detailed in section 6.6.2. This was done to provide additional recommendations for future virtual laboratories and identify weaknesses that could be addressed.

3.3.1 BCMB3X01 – Gene and Genome Regulation

Overall, the teaching staff of the first virtual laboratory used were satisfied with result of using the virtual laboratory in their practical class, as it "enabled students to optimise parameters to obtained meaningful results." and "allowed students to organise/optimise their experimental design, to run through the experimental protocol multiple times, to explore the consequences of varying each experimental step in a risk-free and rapid manner, and to generate their own data for analysis." Challenged from previous online years were identified as low student engagement and detachment from the data analysed. The instructors felt that these challenges had been addressed, and that overall student feedback was positive.

While instructors acknowledged that the virtual laboratory was not able to teach physical practical skills, both would use the virtual laboratory again in future years.

Recommendations include adjustments to the virtual laboratories interface, more realistic results generation in cases where the protocol was deviated away from, and additional guidance for students within the virtual laboratory.

3.3.2 BCMB3X01 – Biochemistry of Human Disease

The course coordinator for this course also believed that the virtual Western blot laboratory had helped to address challenged in online teaching and provide students with a better education experience. The virtual laboratory also provided students with opportunities for experimentation that would not be possible in a physical setting, such as large deviation from the protocol. If possible, the laboratory course would be used again for both face-to-face and online cohorts as a training tool for students to test out different scenarios. Additionally, the main challenged faced by this cohort in previous online years was student engagement, which was felt to be addressed by the virtual laboratory tool, with student engaging in with the online material to a greater degree because of the interactive use of the virtual laboratory.

3.3.3 BMSC339 – Cellular Regulation

The virtual laboratory used for the BMSC339 course was found by the coordinator to be a good data-generator for students to experiment with. This was the first time this course had been delivered online, so it could not be compared to previous years, but

there were multiple benefits over previous face to face years. These benefits included the ability for students to attempt the protocol as many times as they wished, the quicker speed of running a virtual protocol, the capability to work outside of laboratory hours, and the generation of personalised datasets for analysis. The virtual laboratory could not provide safety, teamwork, or technical skills however, and would be the main benefits of a physical practical. If possible, this course would use the virtual laboratory again for future cohorts.

4 Discussion

4.1 Addressing the gap

Practical courses are core components of many undergraduate STEM courses, and while traditionally have been conducted in physical laboratory spaces, in recent times there has been a shift to online learning. When transitioning a practical course to permit online learning, course coordinators face challenges in maintaining the quality of the education delivered to their students. These challenges include engaging students, teaching them laboratory theory, and relevant practical laboratory skills.

To address these challenges faced by undergraduate biochemistry courses that had been moved online in 2021, three online virtual laboratories were created as replacements for their physical counterparts. The objective of these laboratories was to foster an active learning environment for students that would allow them to engage with the virtual laboratories, conduct the protocol of the laboratory, and generate realistic data based upon their actions within the program.

4.2 The Virtual Laboratories

These virtual laboratories were deployed to over 250 students across the three separate undergraduate courses, acting as substitutes for three distinct immunodetection experiments. Each of the laboratories was developed in the scope of a specific investigation and generated results that students could analyse after completing the experiment. In this way students were able to investigate the aim of their specific laboratory as they would in a physical laboratory and discuss the

biochemical implications of the results they obtained. The process of obtaining this data also tested and reinforced the students' theoretical knowledge of the technique, requiring them to understand the theoretical mechanisms underpinning the technique to succeed in obtaining desirable results. This was because the generation of data was specific to the student, based upon the method that they used to obtain it, and if students had made mistakes in the program, they would acquire accordingly adjusted results. Students could then restart the program after considering why they had acquired the result they did, and reattempt it again immediately.

4.3 User feedback

Student feedback showed that the virtual laboratories were highly enjoyed by students, and that they would also feel more comfortable performing the technique studied in a physical environment after using the virtual laboratory. Students also reported that they did use the tool repeatedly to iteratively improve their output by examining their results obtained and theorising what could be performed differently, a primary goal of the study. Some students did respond, however, that they quickly iterated through these cycles, and tried many options until obtaining a working output. This trial-and-error method is not an optimal learning method, and adaptations should be made in future virtual laboratories to discourage this approach.

All course coordinators of the three laboratory units responded positively to the use of the virtual laboratory over other online content and would consider using it again in future years. Coordinators found that it succeeded in addressing the primary challenges faced when delivering online content, which was identified to be low student engagement in less-interactive online environments. Additionally, the virtual laboratories were found to possess some benefits even over physical laboratories, such as the ability of students to repeat and experiment with the protocol as desired, which is normally prohibited by time and financial constraints. As such, the virtual laboratory could be used alongside a face-to-face practical laboratory course, to allow students to familiarise themselves with the protocol before participating in the physical version.

Virtual laboratories were not considered to be suitable as a complete replacement for physical laboratories in the opinion of students and course coordinators. The downsides identified included being unable to teach physical practical skills such as pipetting, or operation of laboratory equipment, which were identified by both students and instructors to be the primary benefit that physical laboratories possess.

4.4 Areas for improvement

When students and instructors were asked to identify potential improvements to the virtual laboratory, the most commonly suggested changes included improved graphics, additional opportunities for students to make consequential decisions, and a more detailed accompanying instruction sheet to guide them through the process. All of these suggestions were targeted improvements that could be implemented into the virtual laboratory, rather than suggesting a necessary return to physical laboratory space, and with additional development time could be incorporated into the programs.

In future, it is recommended to continue the uptake of virtual laboratories into university courses as teaching tools, as they have been demonstrated to have the capability to act as valuable educational tools for students. Courses that would otherwise use laboratory recordings or other passive online material are highly recommended to adopt virtual laboratories, as the student experience is vastly improved. In face-to-face courses, virtual laboratories can provide useful training tool and companion to physical laboratories, and it is recommended to use these tools as a supplement to the laboratory experience.

4.5 Conclusion

The study was successful in creating and implementing three instances of data generating virtual laboratory tools that allowed students to engage meaningfully with online practical coursework. Learners and teaching staff alike found the virtual laboratories to be valuable educational resources and recommended their further use in practical courses.

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